a) Title page BMBF Förderung "Moderne Züchtungsforschung für klima- und standortangepasste Nutzpflanzen von morgen"

Title: Co-Pan-Genome: Computational inference of GxGxE interactions to identify climate-change resilient pathogen resistance in barley. BarleyCOPA

Vorhabenthema auf Deutsch einzutragen: Co-Pan-Genom: Computergestützte Ableitung von GxGxE-Wechselwirkungen zur Identifizierung klimaresistenter Pathogenresistenzen bei Gerste

Forschungsthemen:

a) Anpassung von Nutzpflanzen an abiotische und biotische Stressoren,

b) Erhöhung von Ressourcennutzungseffizienz und Nachhaltigkeit,

c) klimarobuste Etablierung neuer Merkmale („Traits") beziehungsweise Funktionalitäten,

d) Erhalt beziehungsweise Zunahme von Agrobiodiversität und Bodengesundheit sowie

e) Einbindung modernster Züchtungs-Bioinformatik.

Fördermodul: A - Gerste

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b) Abstract

The interaction of crop host genotypes with pathogen genotypes determines resistance of crops and the severity of epidemics by under a given climatic / environmental condition. These Genotype x Genotype x Environmental (GxGxE) interactions determine the resilience of future disease resistance in crops. Modern breeding requires thus new genetic sources of disease resistance robust to pathogen diversity and resilient to changes in environmental factors, which cannot be comprehensively uncovered by classical crop GWAS or QTL mapping which ignore GxGxE interactions. We focus on barley and two of its main (most complex) pathogens *Fusarium graminearum* responsible for the *Fusarium* Head Blight (FHB) also characterized by the production of mycotoxins, and *Ramularia collo-cygni* the agent of Ramularia leaf spot (RLS). Both diseases have become more widespread and causing more yield loss with change in climatic conditions since the 1980s. However, breeding resistance to these pathogens is impeded by the strong effect of GxGxE interactions, meaning that barley genotypes are not consistently resistant to pathogen strains depending on abiotic conditions. Thus, chemical crop protection is often required, which is detrimental for soil health and biodiversity.

Our working hypothesis states that new methods of Genome-to-Genome associations (GtoG or co-GWAS) jointly using barley genetic diversity (SNPs in the barley pan-genome, or Copy Number Variants; CNVs) and pathogen diversity (SNPs and CNVs) across different environments provide a new catalogue of novel resistances, specific or resilient to climatic factors. We build a unique combination of expertise ranging from theoretical evolutionary biology, statistical inference, population (pan)genomics, molecular phytopathology, remote sensing methodology, and plant breeding. In our barleyCOPA project we will derive a novel GtoG framework integrating barley genetic diversity, pathogen genome evolution and epidemiology and climatic data across field locations to reveal the genetic basis of barley x pathogen x Environment interactions. Data will be collected in a unique experimental set-up of five fields across Germany at which 200 sequenced (full genome and SNParray) lines of barley (wild, landraces and modern breeding material) are grown. We collect and sequence over two years more than 1,000 isolates of *Fusarium* and of *Ramularia* across fields. In addition, we are going to conduct a disease severity monitoring by remote sensing methods (drones, hyperspectral cameras) and qPCR quantification over four years at each field for phenotyping and to assess the importance of climatic factors for disease dynamics. Using our novel GtoG method we will predict genes underpinning barley resistance to given pathogen strains and its robustness to environmental conditions. We will further validate the genetic basis for quantitative resistance and effect of abiotic factors in controlled abiotic conditions using barley and pathogen transcriptomics. These results will provide our private breeding partners with invaluable data on a set of barley varieties across a range of environmental conditions, and also point towards genotypes presenting resilience in resistance to pathogens across environments. These genotypes will be used within the first phase to produce, by crossing, new barley lines with improved resistance to *Fusarium* and/or *Ramularia*. We will produce novel bioinformatics resources, high amounts of genomic information, scientific knowledge, and advanced barley lines for crop production. We will generate benefits for biodiversity and crop resource efficiency by providing tools to replace chemical plant protection by climate-resilient, disease resistant barley.

c) State of the art

Breeding for disease resistance in crops is an arms race between breeders and pathogens (1, 2), accelerated and complicated by climate change. Indeed, the susceptibility/resistance of hosts and the severity of epidemics is determined by the interaction of crop host genotypes (variety, lines) with pathogen viruses, bacteria, fungi genotypes (strains, isolates) under a given climatic / environmental conditions. These Genotype x Genotype x Environmental (GxGxE) interactions (2, 3) determine the resilience of future disease resistance in crops. Modern breeding requires thus new genetic sources of disease resistance, robust to pathogen diversity and resilient to changes in environmental factors. However, the classic mapping strategies (QTL mapping or Genome Wide Association Studies, GWAS) to uncover new sources of resistance in crops are hindered by GxGxE interactions (4). With progress in sequencing technology, it is within reach to obtain polymorphism data of plants and their pathogens to uncover GxG interactions determining susceptibility/resistance and virulence/avirulence. Despite the promises of such co-GWAS (Genome-to-Genome, GtoG, (5, 6)) for the future of crop breeding, these methods have only been performed so far on sequences from infected humans (or wild host species) and their pathogens (7–12). The Tellier lab (TUM-PG) has recently developed a model-based theoretical and statistical (Bayesian) GtoG framework integrating full genome data of HCV (Hepathitis C Virus), infected and uninfected European human individuals. Performing an analysis of all 68×10^6 pairs of human SNP x HCV SNP, 535 GxG associations are found to be key determinants for the infection such as new viral virulences targeting human susceptibility genes and an unknown resistance gene to HCV(13) (Figure 1).

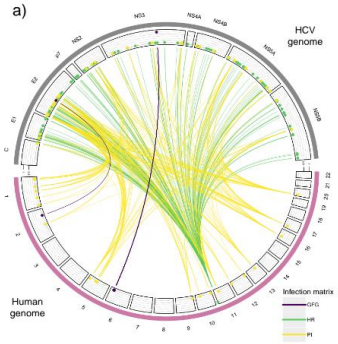

***Figure 1**: Graphical output of the GtoG analysis and the uncovered 535 significant GxG associations coloured by the inferred characteristics of the interaction (Gene-For-Gene in purple, host resistance in green, pathogen virulence in yellow). The human chromosomes are shown on the bottom, the virus contigs on the top. Figure from (13) .*

The overarching objectives of our barleyCOPA project are 1) to develop and conduct the first GtoG method for crop breeding (mimicking the output of Figure 1), and 2) breed for novel barley resistances robust to GxGxE interactions against *Fusarium* Head Blight (FHB) and *Ramularia* Leaf Spot (RLS) diseases. We focus on spring barley and its *Fusarium* and *Ramularia* fungal pathogens, because with climate change these two diseases are major threats to barley production in Germany and Europe. Indeed, breeding efficiency is limited for resistance to FHB and RLS in barley by the strong effect of abiotic factors (GxGxE) on the prevalence and infectivity of these fungi. We design an innovative experimental and sequencing plan to 1) decipher the genetic bases of GxGxE interactions, 2) uncover novel genes of quantitative resistance and their interacting pathogen genes, 3) validate the function of these barley genes, and 4) ultimately breed new resistant and climate resilient barley varieties with our private partners.

Fusarium head blight (FHB) is caused by a complex of diverse *Fusarium* species and intra-species chemotypes. Chemotypes are defined strains or isolates that differ, among other factors, in the class (*e.g.* trichothecenes or cyclodepsipeptides) and individual derivates of produced mycotoxins (14, 15), which represent a major issue for downstream food processing of barley grains (16). Barley FHB resistance is a quantitative trait inherited via several loci with mostly unknown functions (17), although some known gene products such as the barley deoxynivalenol (DON) 3-glucosyl transferase UGT13248 (18) appear to metabolize the fungal toxins. Such general resistance components may, first, drive pathogen adaptation by selecting species or chemotypes that produce a variability of toxins, such as cyclodepsipeptides, NX-trichothecenes or 3-acetyl DON that cannot be metabolized by UGT13248. Second, FHB and the production of mycotoxins are both greatly influenced by climate and local weather conditions (air humidity, precipitation and temperature (19, 20)) especially before and around anthesis (21, 22). Third, measurements of kernel load with DNA from diverse *Fusarium spp*. in archive barley samples over more than 50 years by TUM-PP reveal a general increase in contamination, in particular for *Fusarium graminearum* since the 1980s. The general temperature increase over years only partially explains this trend (23), and TUM-PP demonstrated a significant effect of host genotypes on fungal DNA and mycotoxin loads (22). TUM-ENI/TUM-PP, in collaboration with CAU-PP (24), have further shown that drought can support or suppress FHB (25), and RNAseq analysis uncover 18 different barley gene co-expression modules. Infection is thereby linked to the DON detoxification *UGT13248* gene as a hub in a major coregulon (25). Together, these results support the importance of pathogen genotype (chemotype) x weather x host genotype GxGxE interactions in determining barley FHB severity. We focus during the first phase of the project on *Fusarium graminearum* which populations exhibit high genetic diversity and recombination and little geographic pattern of isolation (at the scale of Germany). However, some chemotypes might locally enrich or newly emerge over time (26, 27) generating population clustering maintained by selection (28). Hence, understanding the dynamics and genomic/genetic bases of *F. graminearum* adaptation to barley varieties in the German agricultural landscape by accounting for GxGxE interactions, is crucial for the design of future FHB resistance breeding programs.

Ramularia collo-cygni (Rcc) the causal agent of Ramularia Leaf Spot (RLS) has become since the 1980s a global major barley pathogen, exhibiting resistance to major fungicides and causing up to 25% of yield losses (29). While Rcc can be detected early in the season (in plants and grains), the pathogen initially grows often symptomless and thus is difficult to detect. Therefore, improvement on the speed and efficacy of disease quantification in the field is essential as demonstrated by work at HSWT (Prof. Noack) with remote sensing and automated quantification of barley yield (30–33). Joint work by TUM-PP and LfL demonstrated quantitative variation in RLS, as well as FHB, severity between 15 major cultivars (including cultivars from commercial breeders) and strong effect of long-lasting drought in reducing disease development (34, 35). It is thus hypothesized that a GxGxE combination of altered physiological properties in modern barley lines, climate change, but also an extended growing season lead to increased host susceptibility (34, 35). Despite being described as a clonal pathogen, Rcc in fact exhibits extensive genetic diversity and recent population expansion (36). In this joint work by CAU-PP, TUM-PP and TUM-PG, we generated a new genome assembly as well as a small scale pan-genomic resources for *Ramularia collo-cygni* (36, 37) serving as starting point for the current proposal. Hence, understanding the dynamics and genomic/genetic bases of Rcc adaptation to barley varieties and abiotic conditions in the German agricultural landscape is crucial for the design of RLS resistance breeding programs.

The genome sequencing and subsequent pan-genome projects (38–40), led for the bioinformatics analysis by H-PGSB (Dr. Spannagl), represent milestones for barley breeding. These resources are invaluable for the breeders to develop novel varieties of barley as done by LfL and HSWT (41, 42). The bioinformatics tools and know-how from H-PGSB (43, 44) allow us as a consortium to leverage genomic information on pan-genomes and Copy Number Variant (CNVs, gene presence-absence between individuals) diversity in a large panel of barley lines ranging from wild, landraces up to modern breeding material (from our private partners). Indeed, coevolution between plants and pathogens is thought to occur by fast birth and death process at genes involved in the interactions (45). There is evidence in *A. thaliana* for large copy number variation at resistance genes across populations (46), while, as shown by CAU-PP and TUM-PG in wild tomatoes, most resistance genes belong to the pan-genome (are present in all individuals) but exhibit large nucleotide diversity in the coding regions (47). Therefore, analysing the full genome sequences for nucleotide diversity in the pan-genome (SNPs), but also for presence-absence diversity (CNVs and gene families) in barley as well as *Fusarium* and *Ramularia* genomes is key to uncover the genes involved in GxGxE interactions. Functional validation of these genes can be conducted by measuring gene expression in response to infection and abiotic factors and understanding their co-regulated environment. For instance, TUM-PP and TUM-ENI have analysed with H-PGSB barley transcriptomic response during powdery mildew or *Fusarium* infection, and thus identified among others, barley disease susceptibility genes (48) and mycotoxin-response genes underpinning potential GxG interactions (25).

The barleyCOPA team presents a unique combination of expertise to design an innovative GtoG based methodology for next generation plant breeding. Dr. Herz is responsible for barley breeding research at LfL with a focus on resistance against pathogens (RLS in particular) and conducts field work across Bavaria to uncover the effect of environmental conditions on disease severity. The Hückelhoven lab (TUM-PP) studies plant–pathogen interactions across scales, using cell biology, reverse genetics (including barley CRISPR technology) and omics techniques, to understand the genetic underpinnings of disease resistance and susceptibility in the field, and under controlled conditions, in pathosystems of socioeconomic relevance. The Noack lab (HSWT) is developing remote sensing techniques to monitor diseases of cereals in the field, and predict crop yield. Dr. Spannagl (H-PGSB) develops novel bioinformatics tools for analysing and assembling genomes and pan-gemomes of crops such as barley, wheat or maize. The Stam lab (CAU-PP) investigates the genetic and genomic diversity and evolution of plants and their pathogens in diverse plant-pathosystems (*Fusarium spp*. and *Alternaria solani* on potato and tomato, *Ramularia collo-cygni* on barley). Dr. Steidele (TUM-ENI) is a senior postdoc and molecular plant pathologist, who is studying plant pattern-triggered immunity and transcriptional disease resistance gene networks (25). She currently establishes tissue culture and transformation tools for targeted mutagenesis and over-expression in barley. The Tellier lab (TUM-PG) builds mathematical models of coevolution between hosts and pathogens and develops ad hoc statistical methods for GtoG analyses using omics data. Prof. Wespel (HSWT) is responsible for optimizing established plant breeding schemes, selection strategies and plant phenotyping using new technologies. She worked previously as a spring barley breeder to adapt barley to climatic and biotic stresses, and optimizing malting and brewing quality. Therefore, our project barleyCOPA falls squarely into the objectives of the BMBF call for modern breeding methodology because we combine the expertise in bioinformatics, omics analysis, experimental work on the barley - *Fusarium/Ramularia* pathosystems, a proven history of successful collaborations between TUM-PG, TUM-PP, TUM-ENI, CAU-PP, LfL and H-PGSB and contribution by HSWT for the remote sensing technology and

development of new breeding programs with our commercial partners SZ Ackermann, SZ Streng-Engelen. SZ Ackermann and SZ Streng-Engelen are, and have been, involved in successful bi- or trilateral collaborations with our consortium partners (TUM-PP, TUM-ENI, LfL, HSWT).

d) Project objectives

The overarching and ambitious objective of this proposal is to develop and conduct the first GtoG method for crop breeding starting from the design and development of novel mathematical model, statistical method and bioinformatics tools, up to the translation into new barley varieties with improved resistance to *Fusarium* and *Ramularia* and climate resilience. Both pathogens currently require the use of chemical crop protection to avoid yield losses, which is harmful for soil agrobiodiversity and not wanted in the light of the EU GREEN DEAL. Our project, barleyCOPA, encompasses the design of new experimental procedures and conducting field trials with 200 barley lines across five locations (three by public and two by the private partners), sampling and sequencing of all barley lines as well as over 1,000 strains of *Fusarium* and of *Ramularia* over two years, monitoring of disease dynamics and yield measures, performing GWAS for resistance, up to the functional validation of the relevant genes and test of their resilience across environments. The aims of barleyCOPA are threefold, and can only be achieved by the unique combination of expertise in theoretical biology, functional and evolutionary genomics, molecular phytopathology, remote sensor technology and crop breeding.

First, we develop a genome to genome (GtoG) framework for crop-pathogen analyses of GxGxE interactions. Such framework will bridge the field of plant disease epidemiology with crop population (evolutionary) genomics. We ambition that the novel bioinformatic and genomics analysis methods can become the new gold standard for future breeding of crops for resistance to pathogens accounting for environmental effects. To achieve this aim, we derive the following operational objectives (WP 1):

- To develop a population genomics model and genome simulator for pathogen populations across fields,
- To build a Susceptible-Exposed-Infected-Removed epidemiological model of disease dynamics per field,
- To design and compute novel indices for GtoG comparisons within each experimental field and build an Approximate Bayesian Computation (ABC) method using these statistics,
- To assess the general statistical power of the GtoG ABC method for the barley-*Fusarium* and barley-*Ramularia* systems, and generally for other crop-pathogen systems,
- To quantify the effect of the environment (climate) on the strength of the GxG associations across fields,
- To develop an experimental design tool for the commercial breeders.

Second, we will conduct a unique set of field trials at five locations (Kiel (CAU-PP), Weidenbach (HSWT), Freising (LfL), Irlbach (Ackermann), and Uffenheim (Streng-Engelen)). These fields reflect different environments in Germany to uncover barley adaptation to abiotic and biotic factors. The operational objectives are (WP1-7):

- To conduct five field trials across Germany, each with 200 barley lines (WP2-5),
- To collect climate data for all the trial locations,
- To sample *Fusarium* and *Ramularia* strains found in these trials, crucially keeping the information of each strain location and host variety of sampling (WP4, 5 and 7) uncovering fungal biodiversity,

- To obtain full genome data of 200 spring barley lines: 70 readily available from an existing pan-genome project, and modern breeding material with 30 sequenced with long read PacBio, and 100 sequenced with short reads Illumina (WP6) as well as genotyped by 50K SNP-array,
- To develop bioinformatics tools to conduct pan-genome (SNPs) and CNVs analyses (WP6),
- To quantify the amount of RLS infection in our fields using remote sensor technology (WP3) and qPCR (WP4) for all four years and collect climatic data per field (WP3),
- To quantify the amount of FHB infection in our fields using remote sensor technology (WP3) and qPCR (WP7) for all four years and collect climatic data per field (WP3),
- To sequence the full genomes of >1,000 strains of *Fusarium* (WP7) and *Ramularia* (WP3) from three fields in year 1 and 2,
- To perform bioinformatics pan-genome analysis (SNPs) and of CNVs for both pathogens (WP 4, 6, 7),
- To perform population genomics analyses of *Fusarium* and *Ramularia* populations (WP4),
- To fit the Susceptible-Exposed-Infected-Removed epidemiological model to each field disease dynamics using data from WP3-5, 7 and climatic data (WP1),
- To perform a barley GWAS for resistance using disease quantification in five fields over four years (WP7),
- To perform GtoG analyses (WP1) based on the Barley (pan-genomes + CNVs) and *Fusarium* (from WP7) or *Ramularia* (from WP4) strains accounting for climatic and epidemiological data (from WP3).

Third, we will generate long lasting contributions of this project such as data management and functional validation of the genetic bases of the barley-*Fusarium* and barley-*Ramularia* interactions (including mycotoxin production) and environmental effects. We will ultimately proceed to develop, with commercial breeders, new lines with stable, durable and resilient resistances across climatic conditions which are thus more resource efficient and improve sustainability of barley production. The improved barley lines with resistance against FHB and RLS will indeed reduce the need for fungicides and can contribute to the maintenance of a healthy fungal soil agrobiodiversity. We have the following operational objectives (WP 8-11):

- To implement a FAIR data sharing system common to all partners with data storage (and database) and GitLab repository for the project and future release of data as well as bioinformatics and statistical software, following the regulations of NDFI (specifically the DataPlant guidelines) (WP8),
- Study the environmental interactions for selected variables (temperature, ozone and CO_2) of the barley FHB pathosystem under controlled conditions (WP9),
- To analyse transcriptomes of chosen barley (eventually CRISPR barley lines or gene insertion mutants in phase II) and newly isolated *Fusarium* strains (and knock out, gene insertion mutants, phase II) in controlled conditions to explain resistance and susceptibility (WP7, 9),
- To develop and perform *Ramularia* inoculation protocols (in controlled conditions) on targeted barley genotypes to confirm field resistance and as a method for secure of phenotyping genetic resources (WP10),
- To perform crossings of promising lines to develop ad hoc novel barley varieties with commercial breeders (Ackermann and Streng-Engelen, WP11). These lines will be screened under controlled conditions and used in new field experiments and in the second phase of the project.

e) Timeliness and originality

To increase resistance of crops (barley) to disease (*Fusarium* Head Blight and *Ramularia* Leaf Spot) and its resilience to changes in abiotic factors, there is a need to uncover new resistance genes by exploring barley diversity. Furthermore, there is also the need to assess the available effect of pathogen diversity on overcoming such possible new resistances, as well as the effect of the abiotic environment on the effectiveness of the resistance. Such knowledge of GxGxE interactions is crucial to optimally deploy resistance varieties in space and in time across the agricultural landscape, with the future aim to be able to predict the effect of pathogen genetic diversity, local climatic conditions and future climate change on GxGxE interactions for barley in Germany and Europe. Such novel durable and resilient resistances across climatic conditions will be more resource efficient, require less fungicide application (preserving beneficial microbes and agrobiodiversity) and thus improve sustainability of barley production in Germany and Europe.

We focus on barley and its two major pathogens which are difficult to control due to their evolutionary potential to evolve new virulences and resistance to fungicides: *Fusarium graminearum* which produces mycotoxins despite partially symptomless infections, and *Ramularia collo-cygni* which produces large damages which vary according to environmental conditions. We will leverage information from full genome data of 200 barley lines, and >1,000 isolates of *Fusarium* and *Ramularia* to decipher the genetic and environmental bases of disease severity in space and time (across fields, years and cultivars). The project is timely as modern breeding tends to erode the genetic diversity of most major crops, and climate change is enhancing the severity and unpredictability of disease epidemics. It is also particularly timely now to take advantage of new sequencing technologies which allow to obtain reveal genetic diversity of full pan-genomes and copy number variants (gene families) for a large number of barley lines and multiple pathogen strains (>1,000 *Fusarium* and *Ramularia*) across multiple fields. It is also timely to develop new bioinformatic, modelling and statistical framework to enhance the field of GWAS and co-GWAS for crop breeding, and to gain new insights from the large amount of data generated using omics technologies. Our project is unique and in its ambition and scale as it combines:

1) the development of a novel Genome-to-Genome framework using jointly genomes of barley and its pathogens,

2) the development of novel bioinformatics tools for pan-genome and Copy Number Variant (CNV) analysis in barley and its pathogens,

3) a new experimental procedure to test for GxGxE interactions allowing to unleash the power of GtoG analyses for crop breeding,

4) the remarkably detailed quantification of infection by *Fusarium* and *Ramularia* each year and across five fields with remote sensing and qPCR quantification,

5) the integration of epidemiological data in a mathematical model for several fields across Germany covering different environmental (climatic) conditions as input for our GtoG method,

6) the functional validation in the field and controlled conditions of the key genes of interaction between barley and *Fusarium/Ramularia* using omics technology, and

7) the development of new barley lines in collaboration with the commercial breeders who contribute a large set of their own breeding material and are conducting field trials at two locations.

This project is therefore only possible due to the inter-disciplinary collaboration between the project partners TUM (Population Genetics, Phytopathology, Environmental Interactions), CAU Kiel (Plant Pathology and genomics), Helmoltz-Zentrum (Plant Genomics and System Biology), LfL (Barley breeding), HSWT (Plant breeding and Information Technology in Agriculture), and commercial breeders (Ackermann, Streng-Engelen). We are thus in the ideal position to bridge the gap between new fundamental research tools and results, and their implementation via the development of new barley varieties by breeders.

Our novel approach integrates three traditionally distinct different fields of plant pathology: population genomics, epidemiology and molecular biology. We thus address challenges in understanding the adaptation of barley to abiotic conditions and biotic stresses, document and quantify available diversity in barley (and its fungal pathogens) by developing beyond the state of the art advanced statistical, modelling and bioinformatic methods. We will ultimately contribute to improve the resource usage and growth of barley plants by providing new pathogen resistance resilient to changes in climatic conditions (and the understanding of the underlying molecular function). The results of our project are far reaching and will benefit many crops beyond barley, contributing to food security and consumer health in Germany and the EU. Our project is rooted in the century long tradition of statistical and mathematical developments at the heart of plant breeding (QTL mapping, GWAS, genomic selection) which encompass the use of novel sequencing technologies to open new areas of progress: in barleyCOPA we integrate the genomic diversity of crop pathogens in the crop GWAS framework.

f) Structure of the consortium and workflow

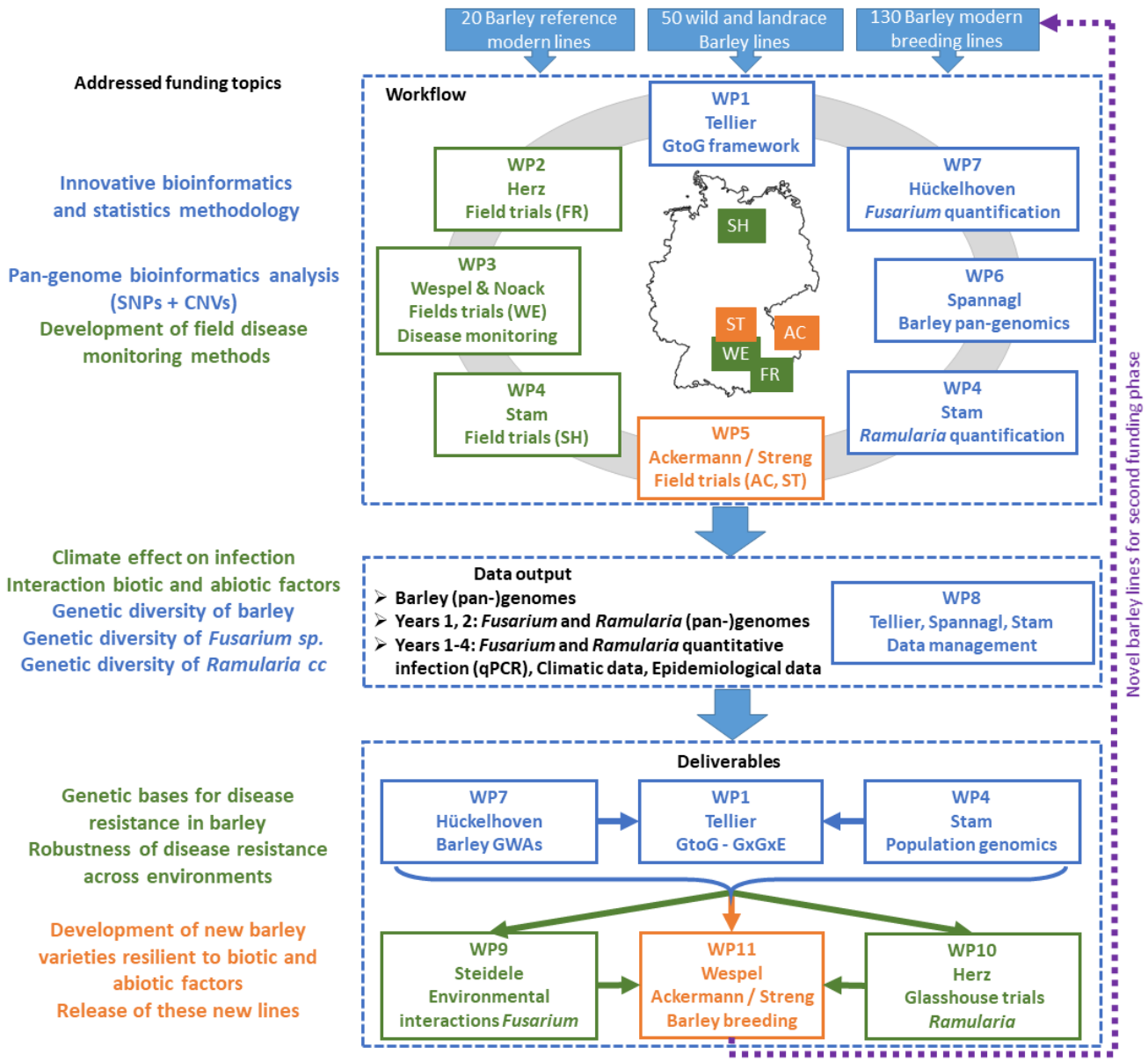

Figure 2: *Summary of the structure of the consortium as well as workflow between workpackages: green for empirical tasks in the field and glasshouse, blue for qPCR, bioinformatics and GWAS tasks, and orange for breeding and generation of novel lines tasks. We have five field locations: SH for CAU Kiel (Schleswig-Holstein), WE for Weidenbach (HSWT), FR for LfL in Freising, ST for Streng-Engelen SZ GmbH &co. KG (in Uffenheim) and AC for Ackermann SZ GmbH & Co. KG (in Irlbach). Data and sample flows are in Figure 4.*

g) Workpackages and financial plan of individual projects

Applicant: Aurélien Tellier (TUM-PG)

Prof. Tellier (TUM-PG) is involved in three workpackages and the financial oversight of the consortium.

WP1: Building and applying the theoretical GtoG framework at the landscape level (GxGxE)

WP1 consists of two tasks, one conducted by a post-doctoral fellow and the second by a doctoral student.

WP1.1: Development of a model-based GtoG method (post-doc)

The novel analysis framework for Genome-to-Genome (GtoG) analysis is composed of three modules (Figure WP1.1) and builds upon the modelling and statistical inference expertise of the Tellier lab in host(plant)-pathogen coevolution and population genomics (e.g. (5, 13, 49–52)).

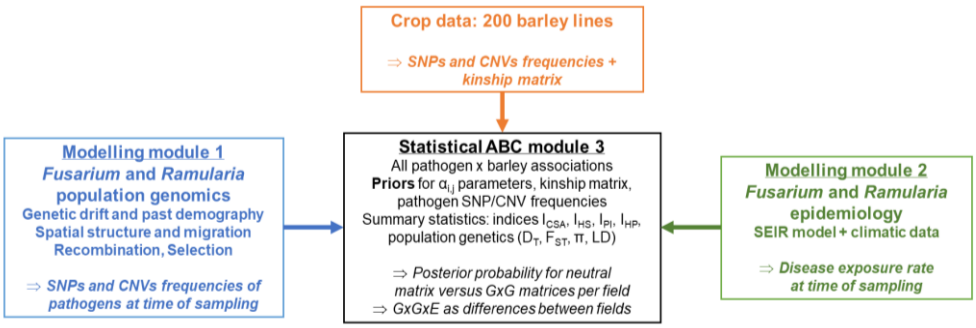

Figure 3: *Schematic representation of the three modules of the barley – pathogen GtoG framework. Polymorphism data from 200 barley lines is used as input. The population genomics module 1 simulates the evolution of full genomes of Fusarium or Ramularia (blue). The epidemiological module 2 generates epidemiological dynamics per field using climatic data (green). The ABC statistical inference module 3 takes as input simulations from modules 1 and 2 and output significant GxG associations.*

The interaction between one or several crop locus/loci and pathogen locus/loci determines the outcome of infection and severity of disease (9–12). The infection matrix (α_{ij}) summarizes the pairwise outcome of the interaction between the ith host genotype with the jth pathogen genotype at two interacting loci with quantitative values of the parameter δ ranging between 0 (full resistance) and 1 (full susceptibility). Similarly to GWAS/co-GWAS (4, 5), the infection matrix quantifies here the contribution of each polymorphic bi-allelic locus (a Single Nucleotide Polymorphism, SNP; or a Copy Number Variant, CNV) to the outcome and severity of infection. Common types of GxG interactions in crops are (53–56):

Crop resistance	Pathogen virulence	Gene-For-Gene (GFG)	Matching-Allele (MA)	Neutral matrix
$\begin{pmatrix} 1 & 1 \\ \delta & \delta \end{pmatrix}$	$\begin{pmatrix} 1 & \delta \\ 1 & \delta \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ \delta & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & \delta \\ \delta & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

Table 1: *Overview of the main interaction matrices for crop pathogen interactions. The neutral matrix represents the null hypothesis, namely these two loci do not determine the outcome of infection. Note that the parameter δ denotes the tolerance of the infection matrix, which accounts for quantitative resistance ($0 \leq \delta \leq 1$). In particular, $\delta=0$ means full total resistance, and $\delta=1$ is full susceptibility.*

In recent work (13), we devised four association indices (I_{CSA}, I_{HS}, I_{PI}, I_{HP}) which are computed for each pair of host and pathogen SNPs (or CNVs) using the observed 1) frequencies f_{ij} of host genotype *i* infected by pathogen genotype *j*, and 2) frequencies f_{iz} of non-infected host genotype *I.*

The first index which we coined as the cross-species association index is defined as $I_{CSA} = |f_{11}f_{22} - f_{12}f_{21}|/\hat{f}_1$ with $\hat{f}_1 = \sqrt{\left((f_{11}+f_{12})(f_{21}+f_{22})(f_{11}+f_{21})(f_{12}+f_{22})\right)}$ being a normalization constant. Note that the I_{CSA} index is analogous to the classic linkage disequilibrium measure used in population genetics and genomics analyses for bi-allelic markers (5, 57).

We derive an index for host susceptibility based on host allele frequency differences between the infected and non-infected host compartment $I_{HS} = |(f_{11}+f_{12})f_{2z} - (f_{21}+f_{22})f_{1z}|/\hat{f}_2$. This index is summarizing information used in classic GWAs for host resistance (58, 59).

We derive an index for pathogen infectivity based on differences in allele frequencies in the pathogen population, in order to reveal the presence of infectivity (virulence in plant pathology) alleles $I_{PI} = |f_{12}f_{22} - f_{11}f_{21}|/\hat{f}_2$. This index captures information used in pathogen GWAs for infectivity (60–62).

Finally we also design an index coined host partitioning, which aims at capturing asymmetry in the infection matrix: $I_{HP} = |f_{12}f_{2z} - f_{21}f_{1z}|/\hat{f}_2$. For example, if the I_{CSA} index allows us to uncover interaction matrices with GFG and MA models, it does not allow alone to conclude which matrix is the correct one between these options. Thus the I_{HP} index is built to disentangle between matrices with symmetrical (MA) from asymmetrical (GFG) values. Note that for the indices I_{HS}, I_{PI}, I_{HP} the normalization factor is: $\hat{f}_2 = (f_{11}+f_{12}+f_{1z})(f_{21}+f_{22}+f_{2z})$.

We then integrated these indices into a GtoG Approximate Bayesian Computation (ABC) method with prior distributions for the unknown (host and pathogen) population genotype frequencies and disease prevalence, and analysed SNP data from 501 non-infected and 451 European humans infected by the HCV virus (and the corresponding virus sequences). From a total of 68x10^6 pairwise GxG (SNPxSNP) associations, 535 were significantly different from the neutral matrix (13)(Figure 1). For this project, we enhance the framework by integrating three different types of data and simulations: population genetics (SNP) diversity data, epidemiological data and disease progression curves, and the GtoG with matrix inference.

Module 1: Module 1 will be developed as a population genomics model allowing to simulate (forward in time with the simulator SLiM (63)) neutral (recombination, genetic drift, past demography, spatial structure and migration) and (positive and purifying) selective events in full genomes of *Fusarium* and *Ramularia.* The parameters will be estimated in WP4 and WP7. Based on these simulations, we will generate a kinship matrix and PCA analysis (PCA components) of the pathogen diversity data. The kinship matrix and PCA components will be used as prior in the ABC (module 3) to account for structure within the *Ramularia* or *Fusarium* samples. These simulations define the ABC priors for the amount of linkage disequilibrium, pathogen genotype frequencies, and sub-structure in the pathogen population entering our fields (Figure 3), so that we can perform a priori statistical power analysis of the optimum number of samples to be sequenced and the likelihood to find relevant loci of interaction and their matrices (Table 1). We will also verify the effect of correcting for clones in the samples of the pathogen based on the effect of the kinship matrix and PCA components on the power of the ABC method.

Module 2: In module 2, we will develop a seasonal SEIR (Susceptible-Infected-Infectious-Removed, (64)) epidemiological (mathematical) model for the *Fusarium* and *Ramularia* diseases. The modelling of disease transmission dynamics in each field yields the probability that a given barley variety receives pathogen inoculum at any point in time (as prior input for the ABC). Using the method (65), our SEIR epidemiological parameters will be obtained from the disease progression measures from WP3, 4 and 7. The fitted epidemiological model is important as it will provide two important information to be included in the ABC in module 3: 1) the spread of disease in a field which defines the rate of disease encounter per barley variety, and 2) the possible genetic relatedness due to the spread of a given fungal clone in our field trials. In other words we want to be able to assess the probability for a barley variety to be infected and by how many spores, which was generically termed disease encounter rate in our work on human disease (13)). Furthermore, we want to model the dispersal of spores between varieties to assess whether several barley varieties are encountering the same pathogen clones. The data of disease monitoring and analysis from WP3 are thus important to fit our SEIR model, and compute these probabilities.

Module 3: We then build the GtoG ABC inference method, using as input data the SNPs or CNVs for 1) the 200 barley lines cultivated as micro-plots in three fields (WP2, 3), and 2) the isolated strains of *Fusarium* and *Ramularia* on each field and each variety (WP4, 7). SNP and pan-genome analysis (CNVs) data will be obtained in WP4, 6 and 7 for *Ramularia*, barley and *Fusarium*, respectively. For each field, we typically generate 10^6 ABC simulations (66) of pairwise barley SNPs (or CNVs) and pathogen SNPs (or CNVs) based on modules 1 and 2. The module 1 simulates pathogen allele frequencies in our fields and population genomics statistics per SNP/CNV (Tajima's D_T , r^2 for LD, nucleotide diversity π, F_{ST} between pairs of fields) based on an *ad hoc* prior distributions (49). Module 2 defines the prior distribution of disease prevalence in a field (13) as well as the structure in the pathogen data. The barley SNPs (CNVs) frequencies are obtained for the 200 barley varieties. Similarly as for the pathogen and as for any GWAs analysis (4-12), we will account for the kinship matrix of the barley accessions in the ABC simulations. In other words, we will use the kinship matrix and PCA obtained from the barley SNP data to control for relatedness between varieties in our ABC. The kinship matrix (and PCA components) will be used as prior to simulate barley genotypes by including the nestedness and modularity of the kinship matrix. These two measures describe the major relationships between varieties.

ABC analyses: We perform 500,000 simulations for each of the 2x2 infection matrices above assuming quantitative resistance values of the parameter (δ) varying between 0 and 1. Using the model choice procedure of the ABC method (66, 67) based on fitting the four indices and population genomics statistics, we infer the type of infection matrix underlying each pair of barley – pathogen bi-allelic loci (SNPs or CNVs). The pairs of loci with a non-neutral infection matrix are the significant GxG interactions using a Bayes factor of 2 against the neutral matrix. We will recover the function and Gene Onthology (GO) of all genes found using the ABC as we may uncover genes relevant for quantitative resistance and general barley response to infection (besides putative major resistance genes).

Extension of the ABC method: We will introduce two extensions of the model of module 3. First we build a diploid host model as some landraces may be diploid and exhibit dominance for resistance/susceptibility. Our interaction matrix becomes thus a 3x2 matrix with a dominance coefficient in addition to the parameter δ. Second, we

will build the model for epistasis between loci as an extension of the 2x2 matrices to higher dimensions. The matrix will typically be of dimension 6x4 assuming a diploid hosts and haploid pathogen with two loci epistasis. This extension is important as for highly polygenic traits it is likely that several loci are involved. All computer codes will be in python and R, integrated in jupyter notebooks and available on our GitLab (WP8).

Deliverables (post-doc):

- 1.1: Development of modules 1-3, assessment of the statistical power of the GtoG inference method (month 24). Yearly parametrization of module 2 (disease data from five fields)(months 9, 21, 33, 45).
- 1.2: Software and modules available on our open GitHub repository (month 24).
- 1.3: Extending the GtoG with epistasis (month 48).
- 1.4: Tool for experimental design of GtoG studies developed with our industry partners (month 36).

Risk management: We do not anticipate major issues in the development of the ABC method or simulation software as the Tellier lab has developed similar tools in the past using python and R libraries (13, 49, 68, 69). We will specifically optimize the simulation software in module 1 to reduce the ABC computational time. Module 2 is based on simple python codes and libraries and a deterministic model, so it should run fast. For module 3 and the full ABC, we will use the LRZ cluster in Munich to run many jobs in parallel. However, we will optimize our codes and pipelines for speed as the number of SNPs in barley and pathogen will likely be large, and several millions of pairwise comparisons need to be performed.

WP1.2: Data analysis of the genomes of barley and Fusarium/Ramularia

Population genomics analyses: In year 1 and 2, the doctoral student will contribute to coding and testing the population genomics simulations of module 1 to include 1) new population genetics statistics (D_T , r^2, π, F_{ST}) in the ABC method, and 2) additional genomics statistics capturing two years of pathogen sampling. We will conduct population genomics (evolutionary) analyses in collaboration with WP4, 6, 7. Using these genetic diversity data per field and across fields, we will parametrize the simulations of Module 1.

GtoG analyses: We will conduct an exhaustive set of GtoG analyses per field and per year: 1) using pan-genome SNPs of barley and *Fusarium/Ramularia* we perform a SNPxSNP analysis, 2) barley CNVs to fungal SNPs, and 3) barley CNVs to fungal CNVs. These analyses will reveal the genetic bases (SNP or CNV) of GxGxE across years and across fields and environments. We hypothesize that under the lack of environmental effects, the same GxG associations should be found across fields, whereas if GxGxE interactions occur, GxG associations are found displaying different matrices across fields (or even no association, neutral matrix, in some fields). When significant GxG associations are found, we will investigate the known genetic bases using available omics data to provide information for further validation and test of environmental interactions (WP9, 10).

Deliverables (doctoral student):

- 1.5: Development of module 1 for population genomics, set-up storage and GitLab (month 24).
- 1.6: Results of GtoG ABC analyses based on genome data from year 1 and 2 (months 28, 40).
- 1.7: GxGxE analyses and set of candidates for WP9 and WP10 (months 36, 48).

Risk management: Based on our exhaustive pathogen strains sampling, high genetic diversity of *Fusarium* and *Ramularia*, and high number of barley lines (200), we will assess whether the disparity in GxG associations across fields is attributed to stochastic sampling of pathogen genotype frequencies or to differential environmental GxGxE effects. If too few infections can be found in our three fields (Lfl, HSWT, CAU Kiel), we can obtain complementary fungal isolates from the two fields from Ackermann and Streng-Engelen.

Outlook for second funding phase: We will add a new module 4 to use our simulations and data to predict future scenarios of pathogen evolution and optimal release of new varieties in an agricultural landscape.

WP6: Barley sequencing

The Tellier lab will perform the DNA extraction for 130 barley varieties using 1) the Qiagen Kit for extraction for short read Illumina, and 2) NewEngland BioLabs kit for long read DNA sequencing. The DNA extraction will be performed by the TA of TUM-PG using leaf material from field and glasshouse experiments at LfL Freising (Dr Herz, WP2). Leaf material is available from varieties already grown at LfL, while other varieties need to be grown in the glass-house for seed production (in 2024 at LfL by Dr Herz, WP2).
Cost of DNA extraction = 950 EUR

After checking the quality and size of our DNA fragments (using our Agilent BioAnalyzer at TUM Freising), we will send DNA for sequencing to external commercial providers. We aim to generate between 10 and 20-fold haploid sequence coverage (65 Gbp) of PACBIO REVIO HiFi data for a total of 30 barley genotypes (requiring 60 lanes of PACBIO REVIO). For the remaining 100 barley genotypes in our experimental setup, we will generate full genome resequencing with Illumina short reads at 20x coverage. Sequencing will be done with a sub-contracting private provider using Illumina NovaSeq (2x150 bp), with one flowcell yielding 360 Gb of clean data, so using 20 flowcells results in approximately 20x genome coverage. For homogeneity of data, we also will apply the 50K SNP-array for barley to our set of 150 modern varieties used in our trials as this is classically used in barley breeding. We can link the SNP found in the array with the full genome data available. The costs for barley DNA sequencing (PacBio long reads, short read Illumina, and 50K SNP-array) are as follows.
Total of sequencing for 200 lines of barley for 290,150 EUR, with 205,500 EUR for PacBio long reads, 77,000 EUR for short read sequencing, and 7,650 EUR for 50K SNP array for 150 varieties.

WP8: Central organization, data management and common platform for bioinformatics

The details of WP8 (shared by the Tellier, Spannagl, Stam and Hückelhoven groups) are in the description of Dr Spannagl project. In a nutshell, we set up a common storage folder and linked GitLab for the barleyCOPA project at the LRZ (Leibniz Rechnungszentrum, Munich) supercomputer facility. We require 100TB of data storage (details below in WP6/8), and this cost of data storage at LRZ are allocated centrally to the Tellier lab.
Costs of data storage at LRZ are 100 EUR / Tb per year so we require 4x100x100 = 40,000 EUR.

The Tellier project is also in charge of centrally organizing a workshop and yearly consortium meeting organization (including status symposium in Potsdam) for a total of 15,100 EUR.

Financial plan TUM-PG

Total personal costs = 533,700 EUR
Total sub-contracting for Barley sequencing = 290,150 EUR
Total for consumables, travel and meetings organization = 78,570 EUR.

Total financial budget for TUM-PG = 902,420 EUR

Projekt Pauschale = 180,484 EUR

Total TUM-PG = 1,082,904 EUR

Applicant: Markus Herz (LfI)

WP2: Field trials for phenotyping at LfL

An assortment of 200 barley lines is to be tested in the field for quantitative resistance to *Ramularia* (RLS), *Fusarium* (FHB) and the locally occurring leaf diseases. Field trials will be performed at locations from CAU Uni Kiel, LfL (Freising) and HSWT (Weidenbach), respectively. In addition, the complete collection of 200 lines will be tested in the field in two locations of the participating plant breeders (Ackermann, Streng-Engelen). The collection consists of reference varieties, including modern, landraces and wild lines contained in the assortment of the pan-genome project (39, 70) as well as breeding lines of the participating breeders (Ackermann, Streng-Engelen) and the LfL. The trials are attached to existing national variety trials or breeding trials. A collection of 200 genotypes including 20 modern reference varieties, 50 landraces and wild barley, and 130 relevant modern breeding lines are cultivated per location and year. This collection of lines is designed to maximize the genetic diversity across lines, namely by including wild and landraces, as well as including elite cultivars from different breeding companies and LfL.

The assortment is evaluated according to the guidelines of the Federal Plant Variety Office for the evaluation of value tests. The TUM and LfL have recognized expertise in the classification of barley for *Ramularia* (RLS). In the Bavarian state variety trials, which are coordinated and supervised by the LfL, RLS is routinely recorded. In addition, remote sensing technology (drones, ultraspectral cameras) from WP3 will be used to score disease progression and severity in each field on each line. Grain samples from the trials are passed on to TUM (TUM-PP) to isolate outgrowing fungi and carry out DNA sequencing on *Fusarium graminearum* (WP7). Infected leaves are harvested directly from the field and sent to CAU Kiel University (CAU-PP) to carry out DNA sequencing on *Ramularia collo-cygni* (WP4). Weather data are available for all test sites and used for the development of prediction (epidemiological) models (WP1). Leaf material from each barley line grown in the greenhouse at the beginning of the project is sent to TUM-PG for DNA extraction and sequencing (WP6, conducted at Prof. Tellier lab using the robot of WP7 in the Hückelhoven lab). Note that we will consult also with the other barley-funded projects to design a set of overlapping varieties which can be used across projects and for which only one sequence will be obtained. Sharing resources, especially full genome data, will enable saving resources and allowing exchange of information and scientists between the funded projects. At the trial sites at LfL, dedicated high-quality measuring stations including sensors for precipitation, air temperature, humidity, wind speed, global radiation and soil moisture are installed close to the field trials. The trials at the breeding stations also have respective measuring stations by the Agrarmeteorlogisches Netzwerk in Bavaria (and run by LfL) nearby. The data regarding the growing conditions are recorded and are documented throughout the seasons.

Deliverables:

- 2.1: Tolerance/resistance measure for all 200 barley lines against RLS in the field trials (months 9, 21, 33, 45).
- 2.2: Climat data for all locations in Bavaria

Risk management: The number of locations is sufficient to obtain the required data and samples in each project year, even if a number of trials fails due to negative climatic conditions or influences which cannot be technically controlled.

#	Genotype	Origin/Breeder	Project
1-70	various	various	Pan-genome Projekt
71	IPZ 24727	Bavarian State Research Center for Agriculture (LfL)	BayKlimaFit
72	Barke	Breun	BayKlimaFit
73	Grace	Ackermann	BayKlimaFit
74	Palmella Blue	NA	BayKlimaFit
75	Marthe	Saaten Union/Nordsaat	BayKlimaFit
76	Morex	Dr. Donald Rasmusson - University of Minnesota	BayKlimaFit
77	Scarlett	Breun	BayKlimaFit
78	Avalon	Breun	BayKlimaFit
79-83	various	various	BayKlimaFit
84 - 87	breeding lines	Ackermann	segregating population 1 Ramularia(extremes)
88 - 91	breeding lines	Ackermann	segregating population 2 Ramularia(extremes)
92	IPZ 24272	Ackermann	donor line Ramularia
93	Sunshine	Ackermann	parental line Ramularia
94	Focus	Ackermann	parental line Ramularia
95	15/729/19	Ackermann	BayKlimaFit (FHB)
96	15/729/73	Ackermann	BayKlimaFit (FHB)
97	15/692/44	Ackermann	BayKlimaFit (FHB)
98	15/692/13	Ackermann	BayKlimaFit (FHB)
99 - 102	22/800	Ackermann	segregating population FHB
103 - 106	22/801	Ackermann	segregating population FHB
107 - 119	breeding lines	Ackermann	pedigree for FHB
120	Gante	Streng	BayKlimaFit
121	Thule	Streng	BayKlimaFit
122	Ven	Streng	BayKlimaFit
123	Tasja	Streng	BayKlimaFit
124	735/18	Streng	BayKlimaFit
125 - 154	breeding lines	Streng	NL-1 (for Ramularia and/or FHB)
155 - 168	breeding lines	LfL	DHL from Rcc Mapping population
167 - 185	breeding lines	LfL	Selected for FHB tolerance
186 - 200	Standards/Elitelines	various	national variety trial (organic and conventional)

Table 2: *Preliminary list of 200 varieties to be used in our field trials with genotype name, origin or breeder and project in which it was developed.*

WP10: RLS disease phenotyping in the greenhouse

In the barleyCOPA project, methods are being developed and improved with which the RLS disease caused by *Ramularia collo cygni* can be investigated in the greenhouse independently of environmental influences. Based on the work of the LfL and the predecessor project „'Gesunde Gerste' – eine biotechnologiegestützte Züchtungsstrategie zur Erhöhung der Widerstandsfähigkeit gegen Ramularia-Blattflecken“ (2009 bis 2011, (71)) existing protocols for artificial infection and infestation detection using reference genotypes are being optimized and established so that they can be used for the phenotypic selection of breeding material. For this project, the tests are set up permanently at the LfL. The inoculum is propagated from fungal spores on optimized growth medium and a defined

number of spores are taken up in water to infect the plants. The barley plants are inoculated after normalization at growth stage 55 to 59. Plants that are only sprayed with water are carried along for control purposes. The plants continue to grow in the greenhouse and are checked for infestation at weekly intervals.

Researchers involved in previous projects are currently employed at LfL. For the proposed project, climate conditions in greenhouses and climate chambers will be optimized and infection and assessment times will be adjusted. The working group's greenhouse was recently renovated and offers the possibility of automatic control of temperature, humidity and lighting. As part of the method optimization, reference varieties are initially screened for RLS. The cultivation of the fungal isolates will be optimized in collaboration with the partners Kiel University (CAU-PP) and TUM-PP for the infection trials in the greenhouse. Once the method has been established in the greenhouse, the entire project assortment of 200 lines will be tested in the greenhouse test. These results will provide a large comparison to the four years of infection collected in the fields by classic scoring as well as remote sensing technologies (WP3). The optimized methods allow the phenotyping of additional genotypes and newly developed breeding lines for the second phase of the project. Phenotyping can be performed quick independent from environmental conditions. We note that the difficulty in studying symptoms and infection of RLS in the greenhouse in controlled conditions has been a major limiting factor for breeding, and we thus aim here to tackle this limitation.

Deliverables:

- 10.1: Greenhouse test methodology for selection of barley for quantitative resistance against RLS (months 36, 45),
- 10.2: Information about correlation between field test and greenhouse test (month 48).

Risk management: There is a low risk that it will not be possible to establish a greenhouse test for RLS screening and selection. All protocols and documentation about the experiments in this project will be made publicly available to help other researchers who want to work on further improvement of the methods. Results from field phenotyping and imaging methods will still provide sufficient information for the most genetic analyses planned in this project.

Outlook for second funding phase: In project years 5 to 8 (second phase), variety prototypes of the Ackermann and Streng-Engelen breeders will be tested in the glasshouse tests and the specific reaction of genotypes to local single isolates of Rcc will be investigated.

Financial plan LfL

Total personal costs = 258,365 EUR
Total for consumables and travel = 43,000 EUR.

Total financial budget for LfL = 301,365 EUR

Projekt Pauschale = 0 EUR

Total LfL = 301,365 EUR

Applicant: Franziska Wespel, Patrick Ole Noack (HSWT)

Work package 3: Field trials and disease monitoring

WP3.1: Field trial for phenotyping

In accordance to WP2 an assortment of 200 barley lines is to be tested in the field for tolerance to *Ramularia* (RLS), *Fusarium* (FHB) and the locally occurring leaf diseases at HSWT. The collection of lines is designed to maximize the genetic diversity across lines, namely by including relevant lines from previous projects (BayKlimaFit and Pan-genome Project), wild and landraces, as well as including lines from different breeding companies and LfL (Table 2). We will conduct a field trial at HSWT in Weidenbach besides the ones at CAU Uni Kiel, and in Freising (LfL). In addition, the complete collection of 200 lines will be tested in the field in two locations of the participating plant breeders (Irlbach (Ackermann), and Uffenheim (Streng-Engelen)). The field trials are carried out according to the guidelines of the Federal Plant Variety Office for conducting official value tests. We have recognized expertise in the classification of barley diseases such as *Ramularia* (RLS) and *Fusarium* and will collaborate with LfL to ensure a similar and comparable scoring according to the official testing guidelines of the federal Plant Variety Office. The same testing guidelines will also be applied to the other field trials at CAU and the breeding stations. In addition, remote sensing technology (see below WP3.2) will be used to score disease progression and severity in each field on each line (micro-plots and yield plots).

Grain samples from the trials are passed on to TUM (TUM-PP) to carry out DNA sequencing on *Fusarium graminearum* (WP7). Infected leaves are harvested directly from the field and sent to CAU Kiel University (CAU-PP) to carry out DNA sequencing on *Ramularia collo-cygni* (WP4). Weather data is available for all trial sites and is used for the development of prediction (epidemiological) models (WP1). For this purpose, climate data is recorded for all field trials using several sensors, including sensors for precipitation, air temperature, humidity, wind speed, global radiation and soil moisture. The trials at the breeding stations also have respective measuring stations nearby. The data regarding the growing conditions is recorded and documented throughout the seasons (according WP2.2).

We will also conduct at HSWT a separate trial with artificial infection with *Fusarium*, to increase the infection pressure and force the occurrence of visible *Fusarium* needed in WP3.2.

Deliverables:

- 3.1: Tolerance/resistance measure for all 200 barley lines in the field trials under documented growing conditions (months 9, 21, 33, 45).

Risk management: The number of locations is sufficient to obtain the required data and samples in each project year, even if a number of trials fails due to negative climatic conditions or influences which cannot be technically controlled.

WP3.2: Disease monitoring by remote and proximal sensing

At all three sites (LfL, HSWT, CAU Kiel) and two breeder's fields, the set of 200 barley lines will be monitored with UAV imagery (RGB + MS) at the relevant growth stages and infestation phases. Additionally, hyperspectral images of the infected plants will be captured at trials sites located in proximity to HSWT and LfL. The collected sensor-based data will be processed creating DOPs and spatially relating the spectral information to scouting and

lab results (30–33). AI models will be built to explain the relation between RGB, MS and hyperspectral data on the one hand and scouting and lab results on the other hand (phenotyping). Our measures by remote sensing will be compared with data from WP4 and WP7 which will measure the quantity of *Fusarium graminearum* and *Ramularia collo-cygni* by qPCR on samples from each of the three fields (CAU Kiel, HSWT, LfL). This will allow to quantify the discrepancy/agreement and how to best use our remote sensing method to estimate the final disease severity. The artificially infected *Fusarium* trial at HSWT (WP3.1) will serve for training AI models. This is necessary, as the natural infection pressure is not reliably high enough in every season to cause strong symptoms of barley FHB.

Deliverables:

- 3.2: AI models for phenotyping based infestation detection (months 24, 36).
- 3.3: Transferability of the results of yield plots to microplots is assessed (month 36).
- 3.4: Information about correlation between remote sensing phenotyping with classical phenotyping and phenotyping by genotyping (qPCR)(month 48).

Risk management: The number of locations is sufficient to obtain the required data and samples in each project year, even if a number of trials fails due to negative climatic conditions or influences which cannot be technically controlled. We have allocated a budget for external services on some sites if the UAV-systems required to collect the data are not available.

Outlook for second funding phase: We will extend the measurements in the second project phase based on the results of the first phase. This will include applying new sensors based on the results from the hyperspectral measurements. We expect these hyperspectral measurements to reveal wavelengths with high significance allowing to identify lower cost multi-spectral devices for detecting plant infestation or even plant infection. We will also use the UAV-data from the first project phase to build multi-seasonal AI models and validate the models with the data collected in the second project phase.

WP11: Barley breeding

In project year two, LfL, HSWT and breeders will perform crosses between relevant resistance donors and elite breeding lines or varieties to begin with the development of improved breeding lines and varieties. The selected progeny of these crosses will be used for validation of the results establishment of the methods for application in the breeding process in the second project phase. Crosses will be made in each year of the project, either as back-crosses, if the genetic distance between crossing partners is large or additional new crosses depending on the information gathered in the progressing project. The field data on the lines' evaluation are incorporated into the GtoG analysis (WP1) and barley GWAs (WP7).

Deliverables:

- 11.1: Set of novel crosses to be tested in the field trials (months 30, 42).

Risk management: LfL and the private breeders (Ackermann and Streng-Engelen) have lines with known resistance, and these can be used for crossing even if few significant GxGxE interactions are found. We use different fields in different environments to maximize our chances to find novel resistances and the associated pervasive abiotic factors.

Financial plan HSWT

Total personal costs = 353,015 EUR
Total other costs for open access publication = 3,000 EUR
Total sub-contracting for field trials and drone flights = 62,840 EUR
Total for consumables, travel and field trials = 27,660 EUR.

Total financial budget for HSWT = 446,515 EUR

Projekt Pauschale = 89,303 EUR

Total HSWT = 535,818 EUR

Applicant: Remco Stam (CAU-PP)

WP4 *Ramularia collo-cygni* sampling and pathogen population genomics

WP4.1: Monitoring Rcc occurrence

To allow classical GWAS for the identification of Rcc resistance QTL and linkage of Rcc occurrence to climatic data, sensitive quantification of Rcc occurrence is needed in each variety. Especially the occurrence of Rcc in the higher leaves is detrimental for the yield and therefore an important breeding trait. Therefore, we will collect 10 leaves per variety at the peak of the growing season (Flag leaf at BBCH 51) for each variety in each plot (2,000 leaves in total). Leaves from the other locations will be collected by the local partners and shipped to Kiel. For each variety the leaves will be pooled per field and DNA extraction for all samples will be done using a Kingfisher Flex liquid handling machine with ZymoBIOMICS 2x MagBeads DNA extraction, optimized for plants. Next, we use two sets of highly specific diagnostic primers (72) to quantify Rcc using RT-PCR relative to the leaf biomass. This sampling will be repeated yearly and the disease severity will be recorded for each barley variety. These data will be shared with TUM-PP for a barley GWAS.

WP4.2: Isolate collection for Rcc

The next objective is to generate a collection for the dominant Rcc isolates from our cultivar collection. For this we will collect leaves with visible symptoms towards the end of the growing season (BBCH 51 - 75). For each barley variety we will collect up to five leaves with visible conidia in each of the 3 field plots. Leaves will be collected by the respective project partners who supervise the fields. Leaf samples will be shipped to Kiel for Rcc isolation using the protocol from (73) with one plate per variety per field. In total we will collect up to 800 (2 fields x 2 replicates x 200 varieties) mixed isolates of *Rcc* per year. Field sampling will be done once in year one and repeated in year 2.

WP4.3: Rcc genome data collection

A central objective of this project is to construct a Rcc genomic diversity database for Rcc to be used in WP4.3 and WP1.2 (TUM-PG). We will grow all isolates (3*200 for year 1 and year 2) in liquid culture in Czapek Dox medium. Mycelia are filtered out, lyophilized and DNA will be extracted using a ZymoBIOMICS 2x MagBeads DNA extraction with our modified protocol optimized for fungi. Sequencing will be done with the DFG Funded Competence Centre for Genome Analysis in Kiel, using Illumina NovaSeq (2x100 bp). One flowcell will yield 360 Gb of clean data, which results in aproximately 20x genome coverage. This is sufficient for all mapping-based follow up analyses (e.g. (36)). We will use the software GATK following the recommended SNP-Calling pipelines (74, 75) to identify genomic diversity per field, per year and between years between all isolates. DNA samples will initially be treated as a pool, thus we allow relax mapping settings and will utilize the function of GATK haplotype caller to resolve dominant pathogen haplotypes. The high sequencing depth will allow this. Data processing will be done as described in WP8. Processing, storage and dissemination of the data will be done in accordance with the Data Management Plan recommendations of NDFI Hub DataPlant.

Link to TUM-PP WP7: Isolate collection for Fusarium and genome data collection

Isolate collection for *Fusarium* will be done by TUM-PP following an identical sampling set-up. TUM-PP will isolate, and pre-process *F. graminearum* isolates for sequencing. Sequencing will be done at CCGA and all data will be processed in parallel to the Rcc data, albeit with optimized SNP-calling and haplotype calling settings, to be determined empirically.

WP4.4: Population genomics structure analyses for Rcc and Fg

To assess the current population structure of Rcc and Fg we will apply population genetics statistical analyses, using a pipeline similar to the one we published before (76) and identify geography or climate-associated differences (or after year 2, year related differences). We will run basic summary statistics analyses (nucleotide diversity π, Tajima's D, F_{ST} between fields) and determine the dominant haplotypes per sample. We also perform classic population structure analysis with STRUCTURE and INSTRUCT as these species also have some clonal reproduction. We perform selection scans (77, 78) to determine genes under positive selection and which exhibit higher migration (or more differentiation) between the locations and over years. Comparison with the results from WP1 will tell us whether these are also the co-interacting loci.

WP4.5 Population dynamics of Rcc and Fg

Next, we calculate population demographic statistics and conduct demographic inference using eSMC (68, 69)to assess the diversity of genetic exchange between the fields. These data will be used in software that can reliably model pathogen migration structure using various methods (79, 80). In a final step we will apply forward simulations to predict pathogen population dynamics using SLIM (63). This information will tell us about future pathogen risks and is required to set up the selection thresholds for genome scans (77, 78) as done in (81) and for optimization of the models of WP1. This is done in collaboration with WP1. These analyses will be done in parallel for Rcc and Fg, but will need parameter optimization for both.

Deliverables:

- 4.1: Quantitative data on Rcc occurrence in four monitored field plots for three years that can be used in GWAS analyses and climatic models (months 12, 24, 36, 48).
- 4.2: Up to 200 Extracted mix-isolates of Ramularia, each representing the infection lesions of from one of the plant genotypes in each of the three fields for 2 years (month 26).
- 4.3: Genomic diversity data for up to 2 * 600 Rcc isolates for WP1 and WP4.4 (months 18, 30).
- 4.4: Population genetics diversity statistics and initial analyses of population diversity (month 39).
- 4.5: Analyses of Rcc and Fg migration and population evolution simulations (month 48).

Risk management: All methods for pathogen isolation, propagation and DNA extraction have been described and used before. Rcc can be readily identified in the field. A potential risk derives from the high number of isolates we aim at. However, in recent other projects we have shown to be able to process 600 pathogen samples in 4 months. Basic SNP calling and population genomics analyses are standard analyses and will yield valuable insights in the

pathogen itself and for other WP. Forward in time modeling of population structure is a high-risk task, and we will collaborate closely with TUM-PG, who have experience with such models.

Outlook for the second funding phase: In the second phase we will continue pathogen sampling. This will amount to an extremely comprehensive genomic data set that can be used for simulations to predict future dominant Rcc and Fg genotypes. Such predictions will be of high importance for breeders and growers.

Financial plan CAU-PP

Total personal costs = 475,700 EUR
Total sub-contracting for Ramularia sequencing = 104,000 EUR
Total for consumables and travel = 77,990 EUR.

Total financial budget for CAU-PP = 657,690 EUR

Projekt Pauschale = 131,538 EUR

Total CAU-PP = 789,228 EUR

Applicants:

Peter Greif (SZ Streng-Engelen)

Isabelle Badasu (SZ Ackermann)

WP5: Field trials and phenotyping for disease resistance

An assortment of 200 barley lines is to be tested in the field for tolerance to *Ramularia* (RLS), *Fusarium* (FHB) and the locally occurring leaf diseases. The complete collection of 200 lines will be tested in two fields. The collection consists of reference varieties, including modern, landraces and wild lines contained in the assortment of the pan-genome project (39, 70) as well as our own breeding lines (Ackermann, Streng-Engelen) as well as lines from the LfL. The trials are attached to existing breeding trials in Irlbach for Ackermann and Uffenheim (Streng-Engelen). A collection of 200 genotypes including 20 modern reference varieties, 50 landraces and wild barley, and 130 relevant modern breeding lines are cultivated per location and year. This collection of lines is designed to maximize the genetic diversity across lines, namely by including wild and landraces, as well as including a diversity of our own breeding material and LfL. The assortment is evaluated according to the guidelines of the Federal Plant Variety Office for the evaluation of value tests. With the help of TUM-PP and LfL with recognized expertise in the classification of barley for *Ramularia* (RLS) we will score RLS disease in each field on each line. In addition, remote sensing technology (drones, ultraspectral cameras) from WP3 (HSWT) will be used to score disease progression and severity in each field on each line. These are invaluable to assess the phenotypes of our breeding lines and uncover potential quantitative resistances in our panel.

With the help from HSWT and TUM-PP, we will sample grain from the trials to be passed on to TUM (TUM-PP) to carry out DNA sequencing on *Fusarium graminearum* (WP7) if samples of *Fusarium* would be needed for sequencing in WP7. Infected leaves are harvested directly from the field and sent to CAU Kiel University (CAU-PP) to carry out DNA sequencing on *Ramularia collo-cygni* (WP4) if samples are also needed to complement the sequencing in WP4. Leaf material from each barley line sent to TUM-PG will be genotyped using the 50K SNP array for barley, and full genome sequencing by long read PacBio or short read sequencing (WP6, conducted at Prof. Tellier lab). Weather data are available from for each field and used for the development of prediction (epidemiological) models (WP1). The trials at the breeding stations have respective measuring stations nearby. The data regarding the growing conditions are recorded and are documented throughout the seasons.

Deliverables:

- 5.1: Tolerance/ quantitative resistance measure for all 200 barley lines against FHB and RLS in the field trials (months 9, 21, 33, 45).

Risk management: Our fields are in locations which differ from those of LfL, CAU Kiel and HSWT, providing additional abiotic conditions for comparison and provide additional samples of naturally infecting *Fusarium* and/or *Ramularia* isolates.

WP11: Barley breeding

From project year two onwards, ourselves, LfL, and HSWT will perform crosses between relevant resistance donors and elite breeding lines or varieties to begin with the development of improved breeding lines. The selected progeny of these crosses will be used for validation of the results, establishment and confirmation of the GtoG methods with view on application in the breeding process in the second project phase. Crosses will be made depending on the information gathered in the GtoG and GWAS analyses. The field data on the lines' evaluation are incorporated into the GtoG analysis (WP1) and barley GWAS (WP7).

Deliverables:

- 11.1: Set of novel crosses to be tested in the field trials (months 30, 42).

Risk management: We have from previous work sets of lines with known resistance, and these can be used for crossing even if few significant GxGxE interactions are found. We use different fields in different environments to maximize our chances to find novel resistances and the associated pervasive abiotic factors.

Financial plan WP5+WP11:

Personalkosten / Personal costs:

- (W) Wiss. Mitarbeiter / Scientific employee: 14468.56 EUR
- (T) Techniker / Technician:11655.20 EUR
- (A) Arbeiter / Worker: 12009.62 EUR

Total personal costs = 38,133.38 EUR

Reisekosten / travel costs:

Participation to the consortium meeting in Kiel

- travel and accommodation for the consortium meeting in Kiel = 235 EUR for W

Participation to the consortium meeting in Freising (2025, 2027, 2028)

- travel for the meeting in Freising = 3x 150 EUR for W = 450 EUR

Total for travel = 685 EUR

SZ Streng-Engelen contributes a total of 76,952 EUR to the project, with the own (KMU) company contribution being 30,781 EUR with a 60% support being asked for the amount of 46,171 EUR.

SZ Ackermann will contribute in kind to the project by conducting one field trial per year, à 25 EUR per line = 20,000 EUR. No funding support is required by SZ Ackermann.

Applicant: Manuel Spannagl (PGSB)

WP6: Barley pan-genomics

WP6.1: Genome sequencing and assembly of barley varieties

Genome sequence assemblies will be generated for all 200 spring barley varieties in the experimental setup of WP2&4. The DNA extraction will be done at the Tellier lab (WP1) and sequencing at a subcontracting partner. A full representation of the gene space with high resolution of (tandem) gene duplications and CNV is essential for the reliable association with phenotypes and traits. The modern breeding genotypes will be selected from the LfL collection of lines as well as from Streng-Engelen and Ackermann sets of lines. We also collaborate with the other funded projects to generate a set of sequenced barley lines to be used across consortia. Indeed, several consortia will also sequence several landraces and modern lines with PacBio. An agreement has been made between consortia to consult with one another and build a list of common varieties to be sequenced to avoid redundant efforts. This coordination is facilitated by Dr Spannagl.

The criterion to choose the lines is to maximize the genetic diversity and that they reflect as much as possible the pedigree structure of the commercial lines. These full genomes will be used to scaffold and guide read mapping and SNP calling of the lines sequenced by short read Illumina (see below). The HiFi reads will be assembled with *hifiasm* (82) and scaffolded into chromosomal pseudomolecules with the help of existing reference genomes from the barley pan-genome. The latest barley pan-genome (70) includes a total of 76 genotypes with full chromosome-scale genome assemblies, comprising wild and domesticated barley as well as landraces. We will use the existing resources for this extensive pan-genome and include 70 respective genotypes in our experimental setup. We thus aim to analyse 100 barley genomes sequenced with long read sequencing technology. In addition, we sequence 100 lines with Illumina short reads. This will allow us to identify potential gene copy number variations (CNVs) with the help of closely related reference genome sequences (from the pan-genome or generated here with HiFi). The key novelty is to constitute a set of high-quality genomes covering the full range of modern breeding lines from the Ackermann and Streng-Engelen commercial partners and others used at LfL for breeding trials.

We will also genotype all 150 modern breeding lines (including those in the latest barley pan-genome project) with the 50K available SNP array to provide adequately formatted data to the breeding partners (Ackermann and Streng-Engelen). These data will also be used to confirm our SNP calls using the short read data or PacBio HiFi genomes, allowing a novel comparison and test of calling accuracy for data from three different sequencing platform. Again, efforts are made with the other barley funded consortia to use several lines overlapping between projects to avoid duplication of sequencing efforts. Dr Spannagl will use our GitLab at LRZ Munich to coordinate the sharing sequence data of interesting barley varieties among consortia.

WP6.2: Barley pan-genome and CNV analysis

The accurate assessment of gene presence/absence and copy number is an essential prerequisite to unlock the full potential of pan-genome data especially in the context of trait discovery for disease quantitative resistance/tolerance. Indeed, it is documented that gene families and paralogs genes do contribute to plant resistances to pathogens, and thus we hypothesize that CNVs may contribute interesting and novel GxGxE interactions. Furthermore, this unprecedented resolution of pan-genome versus CNVs will allow to perform GtoG tests of

novel unknown CNVs pairs of genes. In this work package we will expand on our established pipeline to project gene models in cereal pan-genomes (43, 44, 83) and develop a module capable of accurately calling CNV and (presence-absence variation (PAV) from the all-against-all gene mappings performed therein. This method will have the advantage that no *de novo* gene calling is required, which is time- and resource-consuming. We will use and extend the orthologous gene framework established in the barley pan-genome to classify genes with identified CNV and PAV according to gene family, putative function and phylogeny. This orthologous framework will also build the basis for identifying SNPs within related genes in the extended pan-genome. Alignments of orthologous genes will be queried using the GATK pipeline and resulting SNP calls (69, 70) will be characterised by their containing gene family.

Risk management: The proposed work mainly relies on established and published procedures thus risks are low and specific alternative measures are likely not required. The challenge relies on handling heterogeneous types of data (SNPs from different genomic resources) and vcf files will be carefully check for mismatches due to shift in the gff file of the genome annotation and handling of data format by different bioinformatics tools.

Deliverables:

- 6.1: Sequencing, pan-genome assembly and CNV call for 200 barley lines (months 32).
- 6.2: Software and pipelines for pan-genome assembly available on our open GitHub repository (month 36).

WP6.3: Fusarium graminearum and Ramularia collo-cygni pan-genomes and CNV analysis

The accurate assessment of gene presence/absence and copy number is an essential prerequisite to unlock the full potential of pan-genome data especially in the context of trait discovery for virulence/aggressiveness. Indeed gene families and paralogs genes at virulence and aggressiveness loci are fast evolving and thus do contribute to disease infection and severity (45). Furthermore, this unprecedented resolution of pan-genome versus CNVs will allow to perform GtoG tests of novel unknown CNVs pairs of genes. In this work package we will expand on our established pipeline to project gene models in plant pan-genomes (39, 43, 44, 83) and develop a module capable of accurately calling CNV and PAV from the all-against-all gene mappings from fungi reference genome annotations (37, 84). This method will have the advantage that no *de novo* gene calling is required, which is time- and resource-consuming. We will establish an orthologous gene framework over all *Fusarium* and *Ramularia* genomes using OrthoFinder (85). This will allow us to classify genes with identified CNV and PAV according to gene family, putative function and phylogeny. This orthologous framework will also build the basis for identifying SNPs within related genes in the extended pan-genome. Alignments of orthologous genes will be queried using the GATK pipeline and resulting SNP calls (69, 70) will be characterised by their containing gene family. We thus contribute to help WP4 and WP7 in their assembly of the fungal genomes.

Deliverables:

- 6.3: Sequencing, pan-genome assembly and CNV call for all *Fusarium* and *Ramularia* isolates (months 24).

- 6.4: Software and pipelines for fungal pan-genome assembly available on our open GitHub repository (month 36).

WP8: Central organization, data management and common platform for bioinformatics

In WP8 (shared by the Tellier, Spannagl, Stam and Hückelhoven groups) we set up a common storage folder and linked GitLab for the barleyCOPA project at the LRZ (Leibniz Rechnungszentrum, Munich) supercomputer facility. We require 100TB of data storage linked to the multicore facility of the LRZ for all bioinformatics and genomics analyses. The post-doc and doctoral students of WP1 and WP6 will act as data stewards to organize common data storage and GitLab shared repository between groups. We will provide the integration of DPM (digital project management) including curation of metadata, creation of FAIR digital objects, link computational workflows though GitLab and prepare data for publication though open repositories (ENI, EMBL). We want to implement all pipelines in line with the NDFI regulations (DataPlant guidelines). We will set a GitLab repository for the barleyCOPA project with common folders to access data and private sub-folders for each group/WP. We will use data storage and virtual machine availability at the cloud system of the LRZ in Munich. Hot data needed for analysis are stored in the cloud, while cold data (for storage) are backed-up on the long term storage of the LRZ. All data are linked to a GitLab hosted at LRZ to which all project partners have access. Processed files such as vcf and bam files will be stored for usage by all partners. Raw files will be stored as a backup at LRZ after release on an open-access database upon publication (NCBI and EMBL short-read databases). Note that the processing of raw files will be done separately for each species: Barley by Dr Spannagl, *Fusarium* by Prof Hückelhoven and *Ramularia* by Prof Stam. The resulting bam and vcf files will then be collected on the GitLab for common usage, along with a file describing the variety of sampling of each pathogen isolate. Pipelines for genome-to-genome analysis will be updated and released on the GitLab shared depository.

The coordination of the data analysis and quality standard is done by the Tellier, Spannagl, Hückelhoven and Stam labs. Via the GitLab, sharing of bam and vcf files of barley varieties is also possible with other BMBF funded barley projects. When available via agreement between barley funded consortia, we will also download raw files or bam files from other sequence barley varieties.

Deliverables:

- 8.1: Set-up of storage at LRZ for sharing between groups (month 6).
- 8.2: Software and pipelines for pan-genome assembly available on our open GitHub repository (month 48).

Risk management: We have participated to several DFG TRR and SFB providing data stewardship, and thus will collaborate and advise the post-doc/doctoral student of WP1 for data management and setting up the cluster on the LRZ. The Tellier lab, Stam lab and ourselves use GitLab and jupyter notebook regularly to make our data and codes open access. We will thus also provide teaching sessions for PhD students and post-docs in the barleyCOPA project to use these tools for a more efficient communication and data sharing practice. Dr Spannagl is also in charge of genomic analyses in the other funded barley consortia, so is in an ideal position to centralize information on available datasets and to promote data sharing agreements. During discussions with other funded projects, it was agreed to share information on the sequenced varieties (to avoid duplicating sequencing efforts whenever

possible) as well as to share information on pipelines for read mapping, SNP and CNV calls and pan-genome analyses. Dr Spannagl being in several barley consortia will play a central role to do so. We will use our GitLab to disseminate data and know-how with the other barley funded projects.

Financial plan H-PGSB

Total personal costs = 180,248 EUR
Total for consumables and travel = 48,686 EUR.

Total financial budget for H-PGSB = 228,934 EUR

Total H-PGSB = 228,934 EUR

Applicant: Ralph Hückelhoven (TUM-PP)

WP7: Fusarium Head Blight pathogenesis and GWAS

WP7.1 Phenotyping FHB disease in the panel of barley pan-genome varieties

Barley FHB is often accompanied by high toxin load, even found in symptomless grain material. Therefore, harvested grain from the full panel of barley pangenome genotypes and at least three out of five trial sites with conductive weather conditions (WP2-5) will be used regardless of kernel discoloration or other symptoms (3x200 samples each year). We use each 100 randomly selected kernels per genotype for milling and DNA extraction. *Fusarium graminearum* is expected to be the dominating fungal species from the barley FHB complex (23). We are going to use qPCR specific for *F. graminearum* DNA normalized by barley DNA as a quantitative measure for disease severity and as a proxy for trichothecene toxin load. Quantitative data are considered as pathophenotypes for GWAS (see below). Representative barley grain samples are going to be subjected to LC-MS/MS analysis for confirmation of correlated fungal DNA and trichothecenes contents (22).This will be repeated in four consecutive years of phase one and used to rate FHB resistance in our pan-genome panel. We will identify barley genotypes with robust quantitative FHB resistance or susceptibility over diverse environments and four years in Germany. Data and resistant genotypes can be used by breeders and HSWT for further breeding and by us an TUM-ENI for selection of genotypes.

WP7.2 GWAS-based on pathophenotypes and pan-genome data

We are going to perform genome-wide association studies (GWAS) for quantitative resistance (QR) to FHB and to RLS. For FHB, we are going to use quantitative fungal DNA data from 3 experimental fields from year one, confirmed by data from year two, three and four. Similarly, we are going to use RLS data (both symptoms and qPCR data handed over by CAU-PP, HSWT, LfLand breeders). We will use the barley full pan-genome and CNVs data as a source of genome-wide genetic variation. Therefore, we will use fully assembled and mapped genomes and associated polymorphisms such as SNPs or CNVs (provided by PGSB WP6) as a basis for GWAS on barley QR. This is supported in house by Dr. Marion Müller (https://orcid.org/0000-0001-5594-2319), a trained geneticist who is experienced in GWAS and QTL mapping and at LfL by Dr. Markus Herz. We expect to identify genetic loci for QR to FHB and RLS in spring barley. They will be used for comparison to those loci that derive from the GxG interaction analysis provided by TUM-PG (WP1) and complement or cross-validate those data and to loci that are differentially expressed upon inoculation with *F. graminearum* at TUM-ENI. QR loci are of value for breeders for use as genomic markers or development of molecular markers for FHB resistance.

WP7.3 Isolation of Fusarium fungal strains from individuals of the barley pan-genome set

To provide fungal DNA sequence data for population genomics at CAU-PP (WP4) and GxG analysis at TUM-PG (WP1), we need to isolate fungal strains from individual grain samples. Based on our long experiences with barley FHB, we expect 5-20% of 100 kernels from each of our 600 (3x200) grain samples to harbour *F. graminearum*. For generating single spore isolates of *F. graminearum* from barley grain, each 100 barley kernels per barley genotype with detectable Fg DNA content will be surface-sterilised and put on ¼ potato dextrose agar (PDA) plates. Hyphae growing out from single kernels will be needle-transferred on fresh PDA plates and inspected after

sporulation for characteristic *F. graminearum* fungal conidia morphology. Repetitive spore-dilution series will be used to generate single spore isolates. *F. graminearum* strains will be confirmed by species-specific diagnostic PCR. We are going to isolate at least one fungal strain per barley genotype and site from year one and repeat this in year two (1200 isolates). This will generate important biological material for further fungal DNA extraction and sequencing and for further inoculation experiments (see below). Environmental fungal strains will be maintained as long as data analyses requires and experimentally used strains provided to public strain collections (https://www.dsmz.de/).

WP7.4 Isolation of fungal genomic DNA and full genome short read sequencing

We will isolate fungal genomic DNA from six *F. graminearum* strains per host genoptype (3 sites, two years, 1,200 strains). Sequencing fungal DNA will be carried out according to WP4. In case we cannot reach that number of isolates from our standard procedure of fungal isolation, we´ll enhance the number of kernels per genotype that is used for fungal outgrowth assays. Where appropriate, DNA sequencing resources are combined with *Ramularia collo-cygni* sequencing in WP 4 to share Illumina Novaseq sequencing lanes for short read sequencing. Data will be mapped on the *F. graminearum* reference genome with support from PGSB and CAU-PP and analysis of genomic polymorphisms will be performed by CAU-PP. We will sequence up to 600 strains after the collection in years one and two of the project phase one. Polymorphic regions of the genomes will be manually inspected by us together with CAU-PP for candidate gene functions and manual reannotation, such as effector genes or trichothecene chemotype genes. Data will provide the basis for population genetics (CAU-PP) and GxGxE (TUM-PG) analysis.

WP7.5 Confirmation of host QR and fungal aggressiveness under controlled conditions

To confirm diverse levels of aggressiveness of single *F. graminearum* isolates on single barley genotypes, we are going to use artificial head inoculation of flowering barley in the greenhouse. In our hands, QR to FHB under greenhouse conditions reflects QR in the field (22, 25). To test this also for newly isolated fungal strains and to assess GxG interactions, we will use the following inoculation patterns: **a**) Inoculation of 10 selected barley genotypes, suggested to have diverse levels of QR based on our field data from year one and two, using 10 selected fungal strains from our new isolate collection (4 replicates of 100 inoculations). Fungal strains are selected based on their respective host genotypes (little susceptible versus highly susceptible) and first GxG analyses, *i.e.* differing at loci predicted to be involved in GxG interaction with the selected barley genotypes. We are going to record pathophenotypes by scoring head symptoms and qPCR on fungal DNA in grain samples. **b**) Inoculation of three selected barley genotypes with either single strain isolate inoculum (four isolates) or mixed inoculum from two diverse strains from the same local population (two mixes), which are selected based on our GxG analysis (four replicates of a total of 72 inoculation). We are going to record fungal aggressiveness phenotypes by head symptoms and qPCR targeting genotype-specific fungal DNA sequences (sequence-polymorphism or copy number variation) in grain samples. This may provide evidence for GxG interaction under controlled environments and understanding of fungus x fungus competetive potential. Together data may further support GxG interactions and validate tools for studying GxG interactions (also for phase II, years 5 to 8).

WP7.6 Polymorphic fungal gene expression for understanding fungal aggressiveness

We will use fungal strains, which differ in host genotype-specific aggressiveness according to our GxG analyses, for inoculation experiments to understand differential expression of potential virulence genes. This may be polymorphic genes according to our GxG data or chemotype-specific genes. For the latter, we will establish specific RT-qPCR primer pairs in year one (month 0-6). Therefore, we perform total RNA extraction from three time points after inoculation with four polymorphic strains (24, 72, 120 h post inoculation, hpi, 4 biological replicates) on three host genotypes. Expression of fungal polymorphic and/or chemotype-specific genes is going to be measured via RT-qPCR and normalized to fungal and plant housekeeping gene expression. We chose RT-qPCR over RNAseq here because we can use an informed candidate gene selection and because fungal transcripts are strongly diluted by plant transcripts in complex barley head tissues, which would force us to costly, very deep sequencing. We hypothesize on occurrence differential gene expression at new loci (from GxG) and at chemotype-specific genes in diverse fungal isolates. The latter focusses on genes for general (e.g. *tri5*) and specific trichothecenes such as NIV, DON, 3-acetyl-DON, 15-acetyl-DON, and potentially NX-2, NX-3 genes, which have not yet been evidenced in German *F. graminearum* but in *F. culmorum* isolates (14, 86, 87). For confirmation, the corresponding fungal toxins are evidenced and quantified by LC-MS-MS in the same barley head tissue that is used for gene expression studies. Here, we will focus on specific samples according to differential chemotype expression. Together this will evidence differential gene expression of fungal candidate loci from GxG interaction studies and for chemotype-specific genes. Potential fungal chemotype effects may inform breeders for choosing plant genotypes with potential toxin tolerances.

Deliverables:

- *7.1:* Identified genotypes with quantitative FHB resistance or susceptibility (months 12, 24, 36 & 48).
- *7.2:* GWAS-derived loci for QR to FHB and RLS (month 24 & 36 & 48)
- *7.3:* 1200 host-associated isolates of *F.graminearum* (month 26)
- *7.4:* 2x600 genomes of *F. graminearum* sequenced (months 18 & 30)
- *7.5:* System to study GxG interaction in FHB under controlled conditions. Confirmation of field data for GxG interaction (month 40)
- *7.6:* RT-qPCR methodology for *F. graminearum* chemotype differentiation (month 6). Expression levels of genes in fungal loci from GxG interaction (month 46).

Risk management: All major technologies used here are established at TUM-PP or in the consortium such that we do not expect technological challenges. A potential risk derives from the high number of isolates we aim at. However, due to our experiences, we expect at least 5 out of 100 kernels to bear *F. graminearum*, and if this is overestimated, we could enhance the number of kernels per genotype used for outgrowth assays on PDA medium. In case one of our trial sites would be hardly infected, we can receive samples grain material from the two additional breeders´ experimental sites to get sufficient infection rates.

Outlook for second funding phase: In phase II of this project, we are going to evidence G x G interactions by direct reverse genetic approaches. We are going to establish reverse genetics tools for barley (together with TUM-ENI) and *F. graminearum*. Wild type and genetically manipulated host and pathogen genotypes will be used for FHB

phenotyping and deep RNAseq experiments to understand G x G interaction at the gene expression level of barley (together with TUM-ENI) and *F. graminearum*. If possible, we will include candidate host susceptibility factors into this to generate KO mutants with enhanced FHB resistance.

Applicant: Christina Steidele (TUM-ENI)

WP9: Environmental interactions FHB-Barley

WP9.1 Barley-Fusarium-Environment interactions

Most identified quantitative trait loci (QTLs) against Fusarium Head Blight are highly dependent on environmental conditions (17). It is therefore crucial to study also the environmental interaction of *Fusarium graminearum* and barley. At the Technical University of Munich, we have the opportunity to use TUMmesa, a model ecosystem analyser (https://www.tcf.tum.de/ptc/greenhouses-phytochambers/tummesa/). It features eight walk-in growth chambers where we can precisely control and monitor temperature, humidity, light, CO_2, and ozone. We will study the influence of diverse environmental parameters that may be relevant for pathogenesis, *i.e.* temperature, CO_2 and ozone (O_3), on the barley FHB pathosystem.

We will use the little susceptible (Barke) and highly susceptible (Palmella Blue, PB) cultivars (22, 25) and perform inoculation of flowering barley (growth stage GS 65) with a *Fusarium graminearum (Fg)* strain from our in-house collection or mock inoculation as control. The plants will be grown under the same standard conditions and 2 weeks before flowering (~GS 45), plants will be shifted to the respective environmental conditions or kept at standard conditions for control: higher temperature (30°C), pre-industrial CO_2-concentration or estimated CO_2-concentration in 2075 and higher ozone concentrations. Disease progression will be monitored with visual symptom assessment and head samples will be taken at two different time points chosen based on disease progression. The experimental panel will contain 2 genotypes (Barke/PB) * 2 treatments (mock/Fg) * 5 environmental-conditions * 2 time points * 4 replicates = 160 samples. The experiment will be repeated for validation. To assess the infection rate we will quantify the fungal DNA and measure gene expression for selected barley marker genes (25) and selected *Fusarium* marker genes for toxin production with RT-qPCR, respectively. With support from TUM-PP we will also monitor the fungal in vitro growth rates in corresponding environments.

WP9.2 Validation of observed Barley-Fusarium-Environment interactions

FHB of barley is dependent on weather and thus shows site and year dependent effects. After the first year of field trials, we will select 10 barley varieties representing a broad range of observed susceptibility and phenotypic differences (WP7). Among the selected genotypes, two reference varieties will be included, for which we already have very good data about their susceptibility in the greenhouse (25). Additionally, we will choose the two most frequent isolated *Fusarium graminearum* chemotypes from the field experiments in year 1 (WP7). The 10 selected barley varieties will be grown initially under standard conditions and 2 weeks before flowering (~GS 45) plants will be shifted to climate chambers in *TUMmesa*, which are set to simulate the recorded temperature and humidity of year 1 from two experimental field locations with different FHB severity. We will perform head inoculation of flowering barley (growth stage GS 65) with the two selected *Fusarium graminearum* strains or mock as control. Visual

assessment of the symptoms will be performed at several time points and samples will be frozen in liquid nitrogen for two points in time (10 genotypes * 3 treatments * 2 environments * 2 time points * 4 replicates = 480 samples). We will quantify the fungal DNA and, according to the observed genotype and environment effect, choose 240 samples from which we will sequence the transcriptome (3'RNAseq).

The transcriptome data will be used to identify differentially expressed genes in plants with varying susceptibility levels. Additionally, modules of co-expressed genes will be identified through the application of weighted gene correlation network analysis (WGCNA) (88). These WGCN modules can be statistically associated with so-called external traits, which helps to interpret the transcriptomic data. These external traits are physiological data (like stress marker metabolites) measured during the greenhouse experiment, the visual assessment of symptoms or the quantified amount of fungal DNA. Additionally, binary traits like the genotype or environment can also be statistically associated with the modules (25). Based on the transcriptomic data we will also calculate a gene regulatory network (GRN) (89) to identify key transcription factors (TFs) and their respective targets. Together with the information from the WGCNA, these results are valuable to compare hub genes to genes in identified loci from the GWAS (WP7, TUM-PP) or the GtoG (WP1, TUM-PG) analyses and put them into a broader biological context, or immediately see possible regulatory mechanisms and associated physiological traits.

Identified GRN-derived TF-target gene-pairs will be validated *in vivo* using promoter-transactivation-assays, either by transient Agrobacteria-mediated expression in *Nicotiana benthamiana* or transient expression in barley protoplasts. Five selected and confirmed TFs or other identified candidate genes will be cloned into the respective vectors for gene disruption or overexpression in stably transformed spring barley. The transformation process should be started as soon as the constructs are available to be able to test mutants during the second funding period.

WP9.3 Establishing assays for candidate gene testing

Our long term goal (presumably in phase 2, year 5-8) is to generate stable CRISPR/Cas9-induced knock-out-mutant lines in barley and stable barley overexpression lines to validate the genes identified in the GWAS (WP7, TUM-PP), GxG (WP1, TUM-PG) or the network analysis (TUM-ENI). TUM-ENI currently establishes the protocol for stable Barley transformation using immature barley embryos (90) with first successful transformation events. This method should be then optimised until the end of the first funding period and stable transformed barley lines will be generated for and during the second funding period. Identified barley genes from the GWAS, GxG or network analysis should be initially tested in a transient virus-induced gene-silencing (VIGS) assay in barley spikes to confirm a role *during Fusarium graminearum* infection, before those candidate genes are put in the stable transformation pipeline.

In the first two years of the project, the doctoral student TUM-ENI should establish a protocol for *Barley stripe mosaic virus (BSMV)*-induced gene silencing at the Technical University Munich. To implement this, an Agrobacterium delivery system for BSMV will be utilized. *Nicotiana benthamiana* leaves will be infiltrated to serve as the virus source for secondary BSMV infections. Subsequently, VIGS will be induced in barley spikes through rub-inoculation of flag leaves and spikelets (91) The treated barley heads will then be subjected to infection with *Fusarium graminearum* to assess whether the silencing of the gene-of-interest impacts the severity of the infection

(92). Once the system is successfully established, interesting candidate genes can be tested, before stable transgenic lines are generated or to validate the predicted interactions from the GxG analysis.

Deliverables

- 9.1: Better understanding of the environmental impact on the barley-FHB pathosystem (Month 12). Data can be taken into consideration for interpreting and predicting weather and climate stress impacts on barley FHB (from WP2, 3, 5) and epidemiological modelling in WP1.
- 9.2: Validation of weather effects on barley FHB severity (Month 24). Transcriptomic data for association with loci from GWAS (Month 30). Gene co-expression modules and gene regulatory networks, which help to put identified genes into a biological context (Month 36). Silencing and overexpression constructs for stable barley transformation of up to five selected target genes (Month 48)
- 9.3: Established VIGS-system for validation of candidate genes or predicted GxG interactions (Month 12-24). First set of candidate genes tested for delivery to stable genetic manipulation (Month 48)

Risk management: Together with TUM-PP we have a lot of experience in setting up barley-FHB infection experiments under controlled conditions with additional application of abiotic stress. All greenhouse experiments are planned to be repeated for validation. By selecting only a subset of biological samples for 3' RNA-sequencing following experimental validation through the quantification of fungal DNA, we can significantly mitigate the risk of obtaining non-representative data from the transcriptome.

Outlook for second funding phase: In year 5-8 we would like to validate the found GxGxE Interactions (WP1, TUM-PG) under controlled conditions and test selected candidate genes both in barley and *Fusarium graminearum* knockout or overexpressing mutants in the greenhouse (together with TUM-PP).

Financial plan for TUM-PP and TUM-ENI

Total personal costs = 630,013 EUR
Total sub-contracting for Fusarium sequencing and transcriptomics = 91,660 EUR
Total for consumables and travel = 207,923 EUR.

Total financial budget for TUM-PP-ENI = 929,596 EUR

Projekt Pauschale = 185,919 EUR

Total TUM-PP-ENI = 1,115,515 EUR

h) Coordination of work among partners

The success of barleyCOPA relies on the integration of complementary results and large datasets from the different workpackages WP1-7, which feed into WP9-11. To facilitate the coordination between the different projects tasks, we first devise a data work-flow, and second set-up several feedback and communication loops to facilitate exchange of know-how, ideas and experimental results (and procedure).

1) Data flow and storage

We are committed to implement our project following the best practice of open science and reproducibility (FAIR principle) so all information and metadata are available to all partners for optimal use of analyses. The challenge, and concomitantly the novelty and high-gain part of our project, is in barleyCOPA to integrate a high degree of data heterogeneity to perform new analyses, based on our model-based approach, combining different types of data: several levels of metadata (fields, varieties, years, sample,…), types of data (sequences, SNPs, CNVS, qPCR output, disease severity, transcriptomics, climatic data...) which are obtained redundantly or not by different technologies (sequencing platforms, qPCR, visual scoring, remote sensing,…). Therefore we design a specific workpackage (WP8) to design the data structure and data storage to be used and data stewards (post-doc/doctoral student from WP1) will take charge of the maintenance. A schematic of the samples (barley DNA, *Fusarium* and *Ramularia* isolates and DNA) and data (genome and epidemiological data) is found in Figure 4. We will use integrated DPM (digital project management) with curation and storage of metadata based on the FAIR data usage principle. A central database of all samples and information per field, per variety and per year will be designed and constantly updated. This database will be stored on the project space at LRZ, and shared on the common GitLab repository for access by all partners. Furthermore, we link computational workflows through the GitLab which allows us to share our software and bioinformatics tools for use within the project (version control allows for improvement of pipelines between groups). A public version of the GitLab will then be created to release in open access our software and codes. We will publish the obtained data through an open repository at LRZ and the sequence data of the barley accessions and fungal isolates in *ad hoc* databases (for example EMBL-NBI). We want to implement all pipelines in line with the NDFI regulations, more specifically following the DataPlant guidelines. The GitLab repository and adequate data storage (100 Tb) will be requested at the LRZ Munich, and WP8 is centrally managed by Prof. Tellier, Prof. Stam and Dr. Spannagl (and the post-docs and doctoral students involved in WP1, WP4 and WP6).

2) Flow of ideas and feedback loops

In order to facilitate feedback loops between project workpackages and the flow of information regarding the quality and quantity of data obtained, we will also set up an elements discussion/chat tool for all partners accessing the project GitLab. This allows all group leaders, post-docs and doctoral students to exchange about any change in data structure or measurement undertaken in the field, lab or computer analysis. The changes in metadata or data structure will be implemented in the database. Whenever possible, the committee of doctoral students will be composed of another member of the project for feedback and synchronisation of aims and scheduling. We will meet for yearly project meeting for one day, either ahead of the status-seminar (Plant2030 in Potsdam) in year 1 and 3, in year 2 in Kiel and year 4 in Munich (Freising). We also plan a minimum of one additional meeting per year online with presentations of all WP.

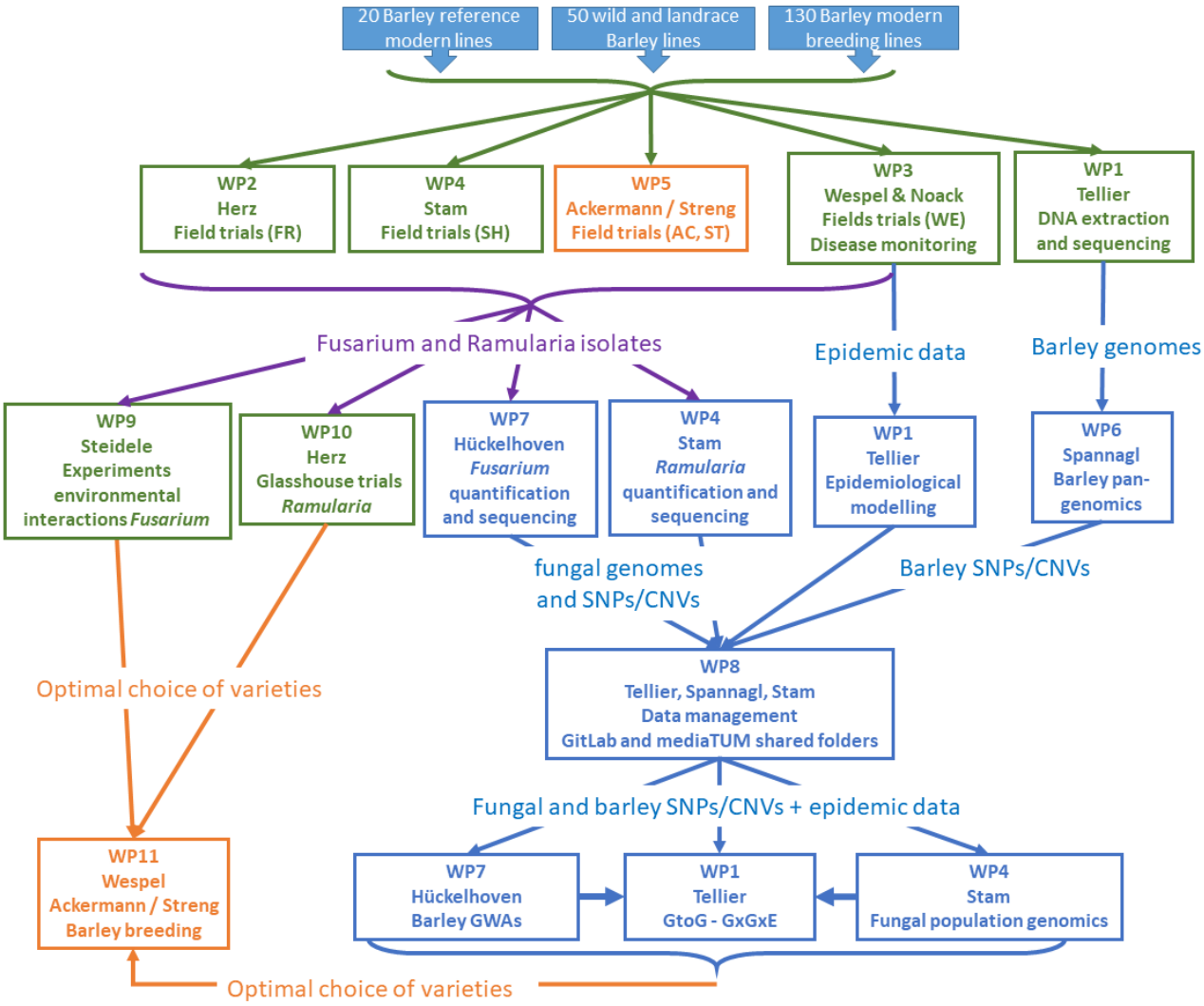

Figure 4: *Summary of the consortium flow of samples (barley in green, Fusarium and Ramularia isoaltes in purple), as well as flow of data in blue (barley genomes, barley SNPs/CNVs, fungal genomes, fungal SNPs/CNVs, epidemiological data). According to the data management plan, data storage and management are performed in WP8 by the Tellier, Spannagl and Stam groups.*

i) Gant chart of deliverables and time schedule

WP	Task	2025				2026				2027				2028			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
WP1	Development GtoG method								1.1								1.3
	Storage, GitLab, codes / method								1.2				1.4				
	Module 1 Pop Genomics								1.5								
	GtoG analyses										1.6		1.7		1.6		1.7
WP2	Field trials			2.1				2.1				2.1				2.1	
	Seed multiplication																
WP3	Field trials			3.1				3.1				3.1				3.1	
	Artificial infection field and AI development							3.2				3.2					
	Analysis infection measures												3.3				3.4
WP4	Monitoring Rcc, sampling, sequencing				4.1			4.3	4.1		4.2	4.3	4.1				4.1
	Population genomics Rcc													4.4			4.4
WP5	Field trials			5.1				5.1				5.1				5.1	
WP6	Barley sequencing, pan-genomics										6.1		6.2				
	Fusarium / Ramularia pan-genomics								6.3				6.4				
WP7	Monitoring FHB, sampling, sequencing				7.1			7.4	7.1	7.3		7.4	7.1				7.1
	GWAS barley resistance to FHB and RLS								7.2				7.2				7.2
	Validation GxG for FHB														7.5		7.6
WP8	Set-up storage and GitLab		8.1														
	Maintenance and improvement of GitLab																8.2
WP9	FHB infections under controlled conditions				9.1												
	Transcriptomics								9.2				9.2				9.2
	VIGS system for validation								9.3								9.3
WP10	Greenhouse test optimization																
	Greenhouse test screening												10.1			10.1	10.2
WP11	Crosses for new varieties and tests										11.1				11.1		

Summary of tasks and deliverables: the colors (green for empirical tasks in the field and glasshouse, blue for qPCR, bioinformatics and GWAS tasks, and orange for breeding and generation of novel lines tasks) correspond to those in the figure (f) for the structure of and workflow within the consortium.

j) Financial plan

Name	Personnel (€)	Subcontracts (€)	Consumables, Equipment and other costs (€)	Sum Costs (€)	Funding rate incl. KMU bonus (%)	Funding (€)	Projekt-pauschale (%)	Projekt-pauschale (€)	Sum Support (€)
CAU-PP	475.700	104.000	77.990	657.690	100	657.690	20	131.538	789.228
TUM-PP-ENI	630.013	91.660	207.923	929.596	100	929.596	20	185.919	1.115.515
LfL	258.365	0	43.000	301.365	100	301.365	0	0	301.365
TUM-PG	533.700	290.150	78.570	902.420	100	902.420	20	180.484	1.082.904
HSWT	353.015	62.840	30.660	446.515	100	446.515	20	89.303	535.818
H-PGSB	180.248	0	48.686	228.934	100	228.934	0	0	228.934
Streng-Engelen	38.133	0	38.818	76.952	60	46.171	0	0	46.171
Sum	**2.469.174**	**548.650**	**525.647**	**3.543.471**		**3.512.691**		**587.244**	**4.099.935**

k) Potential for application and achievements

Wir erwarten, dass barleyCOPA vier langfristige Beiträge liefern wird. Erstens entwickeln wir einen neuartigen theoretischen Rahmen für die Durchführung von GtoG-Analysen, der auf jedes Kulturpflanzen-Pathogen-System anwendbar ist, vorausgesetzt, der Versuchsaufbau ist dem unseren ähnlich. Zweitens werden wir einen Ad-hoc-Verwaltungsplan für die gemeinsame Nutzung von Daten durch die Partner und die Freigabe von Daten bei Veröffentlichungen einführen. Wie bereits erwähnt, sind unsere Praktiken zur gemeinsamen Nutzung von Daten und unsere Datenbank für Daten und Metadaten, die auf den FAIR-Prinzipien basieren, angesichts der Komplexität und Heterogenität der erzeugten Daten, aber auch der gegenseitigen Abhängigkeit der Analysen von verschiedenen Datentypen, wichtige konzeptionelle und praktische Beiträge zu diesem Bereich. Drittens werden wir die genetischen Grundlagen der Gerste-Fusarium- und Gerste-Ramularia-Interaktionen (einschließlich der Mykotoxinproduktion) unter Berücksichtigung von Umwelteinflüssen funktionell validieren. Diese Ergebnisse sind von grundlegendem Interesse für molekulare Pflanzenpathologen. Schließlich werden wir gemeinsam mit den kommerziellen Züchtern im Rahmen des Projekts neue Sorten entwickeln. Dazu integrieren wir 130 moderne Züchtungsmaterialien der privaten Partner Ackermann und Streng-Engelen in unsere zu prüfenden Sorten. Die Daten, die wir über den Krankheitsphänotyp und die Krankheitsquantifizierung (mittels qPCR) erhalten, sind für die Züchter sehr nützlich, um die besten Linien für weitere Kreuzungen auszuwählen. Im WP11 ist geplant, im dritten oder vierten Jahr die ersten Kreuzungen von vielversprechendem Material für weitere Tests zu erzeugen.

1) Errungenschaften: wissenschaftliche Kommunikation

Der in WP1 entwickelte theoretische und statistische Rahmen wird für Pflanzenpathologen und Evolutionsbiologen, die sich mit der Koevolution von Wirt und Parasit befassen, von allgemeinem Interesse sein, so dass wir erwarten, mehrere Veröffentlichungen in hochrangigen Fachzeitschriften (mit offenem Zugang) zu veröffentlichen. In ähnlicher Weise werden die neuen umfangreichen Genomdaten des Gersten-Pan-Genoms und der CNVs eine einzigartige Ressource für die Gerstengemeinschaft darstellen und sollten zu mehreren eigenständigen Veröffentlichungen führen (WP6). Die Arbeitspakete 4, 7 und 9 werden auch grundlegende Einblicke in die Populationsgenomik und den Einfluss der Umwelt auf FHB und RLS liefern und zu einer Reihe wichtiger Veröffentlichungen führen. In WP3 werden wir neuartige Fernsensor-Technologien und Analysen entwickeln, um den Schweregrad von Krankheiten auf Feldern mit Hilfe von Drohnen zu beurteilen. Diese Veröffentlichungen und Ergebnisse sind von zentralem Interesse für Pflanzenzüchter (akademische und industrielle Partner). Darüber hinaus wird erwartet, dass die innovativen Ergebnisse dieses Projekts in Bezug auf die genetischen Grundlagen von GxGxE-Interaktionen und deren Funktionstests in hochrangigen Fachzeitschriften (mit offenem Zugang) veröffentlicht werden. Wir werden dafür sorgen, dass unsere Publikationen durch die Verbreitung in sozialen Medien (X, Mastodon) und über das Kommunikationsbüro des Hans-Eisenman-Forums für Agrarforschung an der TUM bekannt gemacht werden.

2) Errungenschaften: Verbindung zwischen Wissenschaft und Industrie

In WP1 und WP11 werden wir auch Ergebnisse entwickeln, die von unseren Industriepartnern genutzt werden können. Die entwickelte Methode zur Planung von GtoG-Experimenten und Analysen für Industriepartner (Ackermann und Streng-Engelen) wird ein Ergebnis von WP1 sein, das auch als frei zugängliche Creative Common

License veröffentlicht wird. Die Ergebnisse von WP11 im Hinblick auf die Züchtung neuer Sorten werden weiter unten beschrieben.

Darüber hinaus bitten wir um finanzielle Unterstützung für die Organisation eines Workshops im Jahr 4. Dieser Workshop wird unser Projekt sichtbar machen, indem wir eine zweitägige Netzwerkveranstaltung in Freising organisieren, bei der wir unsere GtoG-Methode und unsere Ergebnisse zu Gerste - Fusarium und Gerste - Ramularia vorstellen werden. Wir werden akademische Kollegen aus Deutschland (HSWT, LfL, TUM und Fraunhofer Triesdorf, Uni Hohenheim, IPK Gatersleben) einladen, die an Gerste und anderen Getreidearten arbeiten, sowie Industriepartner (Ackermann, Streng-Engelen), die diese Methode auf andere Kulturen in Bayern (Weizen, Mais und Wintergerste) anwenden könnten. Wir werden auch europäische Kollegen aus dem akademischen Bereich (Prof. McDon-ald, ETH Zürich Schweiz, Dr. Roux, INRAE Toulouse Frankreich,...) und der Pflanzenschutz- und Züchtungsindustrie (Dr. MBoup Corteva AgroScience, Konsortium praktischer Züchter (GFPi),...) einladen, um die Arbeit und Ergebnisse von barleyCOPA zu verbreiten. Wir werden eine neue Generation von Pflanzenzüchtern, Pflanzenpathologen und Computerbiologen für den öffentlichen und privaten Bereich ausbilden und betreuen.

3) Errungenschaften: konkrete Anwendung für die Industrie

Nicht zuletzt besteht ein Hauptziel von barleyCOPA darin, die rasche Herstellung interessanter Kreuzungen aus vielversprechendem Material mit Resistenz gegen Fusarium oder Ramularia und Widerstandsfähigkeit gegen variable klimatische Bedingungen zu ermöglichen. Es ist vorgesehen, dass unsere kommerziellen Partner, Streng-Engelen und Ackermann, ab dem ersten Jahr von einer hochmodernen Phänotypisierung (Fernerkundung per Drohne, Ultra-Spektralkameras) und qPCR-Quantifizierung der Infektion für eine Reihe von 130 modernen Zuchtmaterialien profitieren werden. Da Ackermann und Streng-Engelen ihre eigenen Linien in diesen Versuchen verwenden werden, wird das WP11, falls vielversprechende Sorten auftauchen, für die Durchführung von Kreuzungen zur Entwicklung neuer Linien zuständig sein. Im Idealfall werden diese Linien in der nächsten Projektphase in die nächsten Versuchsrunden einfließen und zu neuen Tests beitragen und können validiert und weiter verbessert werden. Wir stellen uns vor, dass dieses neue Material 1) von Streng-Engelen und Ackermann für den Markt entwickelt wird und 2) in der zweiten Phase des Projekts für weitere Kreuzungen und Validierungen verwendet werden kann. So erreicht unser Projekt sein Ziel, grundlegende neue Fortschritte in der Genomik, Bioinformatik und Modellierung mit der Anwendung zur Entwicklung neuer Gerstensorten zu verbinden, die resistenter gegen Pilzkrankheiten und widerstandsfähiger gegen klimatische Bedingungen sind. Wir heben hier den bedeutenden finanziellen Beitrag von Streng-Engelen und den Sachbeitrag von Ackermann zu unserem Projekt hervor, indem sie Feldversuche durchführen, Proben und Phänotypisierungsmaßnahmen zur Verfügung stellen, ihre Gerstenlinien zur Verfügung stellen und sich für die Entwicklung künftiger neuer Kreuzungen zwischen vielversprechenden Sorten engagieren.

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